

**Finite Dimensional Noetherian
Filtrations & the
McConnell-Robson Question**

Miłosz Szymański

Master of Mathematics with Honours
School of Mathematics
University of Edinburgh
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Abstract

In 1987 McConnell and Robson have posed the seemingly innocent sounding question: does every right Noetherian affine algebra admit a finite dimensional filtration with respect to which the associated graded algebra is right Noetherian? [cf. [MR01] Section 8.3.10]. One of the first counterexamples was given in 1997 by Stephenson and Zhang [SZ97], and so the focus shifted to finding conditions on affine Noetherian algebras such that they admit finite dimensional filtrations which make the associated graded algebra also Noetherian. Partial answers have been given over the years in the works of Stafford and Zhang [SZ03], as well as Greenfeld [Gre23], however the question for the general case remains open.

The aim of this paper is to study a particular affine Noetherian algebra S introduced in the McConnell and Stafford paper [MS89], with the intent of commenting on whether S give a positive or negative answer to the McConnell-Robson question. A secondary goal of this paper is to introduce the Gelfand-Kirillov dimension invariant as well as the ideas of growth of algebras, with the intent being to give a thorough presentation of the Stephenson-Zhang counterexample.

Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

(Miłosz Szymański)

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Notation

In this paper we will use the following notation and conventions. $\mathbf{k}$ denotes an arbitrary field and every algebra considered in this paper, unless otherwise stated, will be a $\mathbf{k}$ -algebra. We will use capital letters from the start of the Latin alphabet A, B, C to denote algebras.

Given a filtered algebra A we have the map $\mathrm{gr}_F : A \rightarrow \mathrm{gr}_F A$. Hence given $f \in A$, we have $\mathrm{gr}_F(f) \in \mathrm{gr}_F A$. However this quickly becomes quite clunky when writing products of elements, hence when there isn't a risk of confusion we will use the short hand $\tilde{f} := \mathrm{gr}_F(f)$ for elements, but use the notation gr_F when we want to stress the map.

The free algebra in n variables over the field $\mathbf{k}$ will be denoted by $\mathbf{k}\langle X_1, \dots, X_n \rangle$ where X_i is a formal symbol for each i .

Given an algebra A , its filtration will often be denoted by $F_\bullet$ or other variants such as $F'_\bullet$, $\tilde{F}_\bullet$, etc. The only exception to this is when $A = \mathbf{k}\langle X_1, \dots, X_n \rangle$ and we will instead denote filtrations on the free algebra by $\mathcal{F}_\bullet$.

We will use $\dim_{\mathbf{k}}$ to denote the dimension of an algebra as a $\mathbf{k}$ -vector space.

In this paper $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_+$ will denote the set of all positive integers.

Given a collection $v_1, \dots, v_t \in A$ of elements of an algebra, we will denote the $\mathbf{k}$ -span of these elements by $\mathbf{k} \cdot (v_1, \dots, v_t)$, and more generally given a subset $X \subseteq A$ of elements, we will denote the $\mathbf{k}$ -span of elements in X by $\mathbf{k} \cdot (X)$. Moreover, the ideal generated by elements $v_1, \dots, v_t$ will be denoted by $(v_1, \dots, v_t)$, and in the case of principle ideals we will also use the notation AvA to denote the ideal generated by $v \in A$, whilst vA and Av will denote respectively right and left ideals generated by v .

Apart from the above remarks, notation will be introduced when it is needed.

Chapter 1

Introduction

In 1987 McConnell and Robson posed the question whether every right Noetherian affine algebra admits a finite dimensional filtration such that the associated graded algebra is right Noetherian [cf. [MR01] Section 8.3.10]. Note that such filtrations are often referred to as a finite dimensional right Noetherian filtrations, and similar naming applies to finite dimensional (left) Noetherian filtrations. Also note that we will from now on refer to the above question as the MR question. Around this time other authors asked similar questions: in 1988 Lorenz asked if every affine (right) Noetherian PI-algebra admits a finite dimensional (right) Noetherian filtration [cf. [Lor88] p.436]; in 1989 McConnell and Stafford asked about what conditions are required on a ring R for it to have a filtration $F_\bullet$ such that $\text{gr}_F R$ is Noetherian or finitely presented [cf. [MS89] Section 3.6]; and in 1999 Yekutieli and Zhang asked whether every affine Noetherian PI-algebra admits a Noetherian connected filtration [cf. [YZ99] Question 6.16].

In 1997 Stephenson and Zhang gave a negative answer to the MR question. In their paper [SZ97], they showed that any graded, locally finite, Noetherian algebra must have subexponential growth. Using this result they then showed as a corollary that any affine Noetherian algebra with exponential growth does not admit a finite dimensional Noetherian filtration, and went on to present an example of an affine Noetherian algebra with exponential growth which provided the negative answer.

In 2003 Stafford and Zhang found counterexamples to the MR question among affine Noetherian PI-algebras, which also give negative answers to the questions raised by Lorenz in 1988 and Yekutieli-Zhang in 1999. With these negative answers, the focus has shifted to finding necessary conditions on affine Noetherian algebras such that they admit finite dimensional Noetherian filtrations. Apart from the results already mentioned, a notable recent result is that of Greenfeld [Gre23] in 2023. In their paper, Greenfeld shows that the Resco-Small algebra (a certain affine, Noetherian algebra which isn't finitely presented) [cf. [RS93]] does not admit finite dimensional Noetherian filtrations. Moreover, Greenfeld shows that there exists a countable field over which there exists a continuum of non-isomorphic affine Noetherian algebras without Noetherian filtrations

but only countably many non-isomorphic affine algebras which admit Noetherian filtrations. As can be seen by the above, many partial results to finding necessary and sufficient conditions to the MR question exist, however the question in the general case remains an open problem.

The main objects of study in this paper are filtered algebras and their associated graded algebras, which we proceed to introduce below. Let A be a $\mathbf{k}$ -algebra. We will now recall certain definitions which will be useful throughout the paper. Most of these may be found in [MR01], stated more generally for arbitrary rings.

Definition 1. A is called *affine* if any of the equivalent conditions hold,

1. A is finitely generated as a $\mathbf{k}$ -algebra, i.e. if for some $n \in \mathbb{N}$, $\exists a_1, \dots, a_n \in A$ such that every $r \in A$ may be written as a non-commuting polynomial in $a_1, \dots, a_n$.
2. $\exists \varphi : \mathbf{k}\langle X_1 \dots X_n \rangle \rightarrow A$, given by $\varphi(X_i) = a_i$ where φ is a surjective algebra morphism, for some $n \in \mathbb{N}$.

The second condition will become quite useful when defining a class of filtrations later on in Section 4.3.

Definition 2. An $\mathbb{N}$ -filtration $F_\bullet$ on A is a $\mathbb{N}$ -indexed family of subspaces $\{F_n\}_{n \in \mathbb{N}}$ of A such that the following conditions hold,

1. $F_n \subseteq F_{n+1}$, for all $n \in \mathbb{N}$,
2. $F_i F_j := \{ab \mid a \in F_i, b \in F_j\} \subseteq F_{i+j}$
3. $1 \in F_0$,
4. $\bigcup_{n \in \mathbb{N}} F_n = A$.

Note that there exists notions of $\mathbb{Z}$ -filtrations and I -filtrations for more general indexing sets, however in this paper we will focus on $\mathbb{N}$ -filtrations. From now on, we will drop the mention of the indexing set and simply use *filtrations* when we mean $\mathbb{N}$ -filtrations. Also note that if $F_\bullet$ is a filtration on A , then we say that A is a filtered algebra with filtration $F_\bullet$.

Definition 3. A filtration $F_\bullet$ on A is finite-dimensional if $\forall n \in \mathbb{N}$, $\dim_{\mathbf{k}}(F_n) < \infty$.

Example 4. Take $A = \mathbf{k}[x]$, then if we define a filtration on A by $F_n := \mathbf{k} \cdot (x^n)$ then we see that $F_\bullet$ is in fact finite-dimensional since $\forall n \in \mathbb{N}$, $\dim_{\mathbf{k}}(F_n) = 1$. On the other hand, if we define a filtration on A by $F_n := \mathbf{k}[x]$ for all $n \in \mathbb{N}$ then we see that $\dim_{\mathbf{k}}(F_0) = \infty$ since $x^i \in F_0$ for all $i \in \mathbb{N}$ and x^i, x^j are linearly independent over $\mathbf{k}$ for $i \neq j$.

Remark 5. The above example serves as a good illustration of the idea that the same algebra can have filtrations which behave wildly differently!

Definition 6. A is graded if there exists an $\mathbb{N}$ -indexed family of subspaces $\{A_n\}_{n \in \mathbb{N}}$ such that $A_n A_m \subseteq A_{n+m}$ and $A = \bigoplus_{n \in \mathbb{N}} A_n$ as $\mathbf{k}$ vector spaces.

Definition 7. If A is graded, then for any $f \in A$ we may write $f = \sum_{i=1}^n \gamma_i f_i$ for some $n \in \mathbb{N}$ where $\gamma_i \in \mathbf{k}$ and $f_i \in A_i$ for all i , and such that $\gamma_n \neq 0$. Then we define the degree of f as $\deg(f) = n$, and we say that f is homogenous if $f = \gamma_i f_i$ for some $\gamma_i \in \mathbf{k}$ and $f_i \in A_i$.

We now show how given a filtered algebra A we can construct a related graded algebra.

Construction 8. Let $F_\bullet$ be a filtration on A , then F_n/F_{n-1} is a quotient of subspaces, and hence another subspace of A for all $n \in \mathbb{N}$. Note that even though F_{-1} isn't defined, it's conventional to set $F_{-1} := 0$ in order to make the notation cleaner. Since the filtration is by definition exhaustive it follows that for any $f \in A$ there exists some $n \in \mathbb{N}$ such that $f \in F_n \setminus F_{n-1}$, we define the F -degree as $\deg_F(f) := \min\{n \in \mathbb{N} \mid f \in F_n\}$. We now define the vector space $\text{gr}_F A := \bigoplus_{i \in \mathbb{N}} F_i/F_{i-1}$, as a direct sum of vector spaces, and define the map $\text{gr}_F : A \rightarrow \text{gr}_F A$ by $\text{gr}_F(f) := f + F_{n-1}$ where $n = \deg_F(f)$ for $f \in A$. Remarkably, we can upgrade this vector space to a $\mathbf{k}$ -algebra by defining the multiplication on homogenous elements $\text{gr}_F(f) \in F_n/F_{n-1}$ and $\text{gr}_F(g) \in F_m/F_{m-1}$ by,

$$\text{gr}_F(f)\text{gr}_F(g) := fg + F_{n+m-1}$$

and extending to general elements by distributivity of addition. We leave it as an exercise to verify that this multiplication is well-defined. $\text{gr}_F A$ is called the **associated graded algebra** of A .

Remark 9. The restriction of the map gr_F to F_n , which we will denote $\text{can}_{F_{n-1}}$, is surjective for all $n \in \mathbb{N}$. That is $\text{can}_{F_{n-1}} : F_n \rightarrow F_n/F_{n-1}$ is surjective for all $n \in \mathbb{N}$, as these are just the canonical projection maps from F_n to the quotient. Crucially gr_F is **not** an $\mathbf{k}$ -algebra morphism, in fact it even fails to be additive!

Example 10. Take $A = \mathbf{k}[x]$, and let $F_\bullet$ be defined by $F_n := \mathbf{k} \cdot (x^n)$. Then on one hand we have that $\text{gr}_F(x^2 + x) = x^2 + x + F_1 = x^2 + F_1 = \overline{x^2}$. On the other hand $\text{gr}_F(x^2) + \text{gr}_F(x) = \overline{x^2} + \bar{x}$. Since $\bar{x} \in F_1/F_0$ whilst $\overline{x^2} \in F_2/F_1$, we see that $\bar{x}$ and $\overline{x^2}$ belong in different graded components and hence $\overline{x^2} \neq \overline{x^2} + \bar{x}$.

Definition 11. A is left (respectively right) Noetherian if any of the equivalent conditions hold:

1. Every left (respectively right) ideal is finitely generated.
2. A satisfies the ascending chain condition, that is any strictly ascending chain of left (respectively right) ideals $I_1 \subset I_2 \subset I_3 \subset \dots$ terminates.
3. Every non-empty set of left (respectively right) ideals of A has a maximal member, with respect to inclusion.

In Chapter 2 we will go over the preliminary background regarding Gelfand-Kirillov dimension and growth of algebras needed to understand the results of the Stephenson-Zhang paper. In Chapter 3 we will discuss the theory behind and present the Stephenson-Zhang counterexample to the McConnell-Robson question. In Chapter 4 we will introduce the algebra S from the McConnell-Stafford paper, and discuss why the results of Stephenson-Zhang, Stafford-Zhang and Greenfeld cannot be applied to S . In Chapter 5 we will present our results on finite dimensional filtrations of S , showing that restricting to a certain class of nicely behaved filtrations the algebra S does not admit any finite dimensional Noetherian filtrations, hence providing a partial answer to the McConnell-Robson question for S . We will discuss the difficulties present in answering this question as well as suggest possible future research directions.

Chapter 2

Growth & Gelfand-Kirillov Dimension

An excellent resource for this chapter is [KL00], many results here can be found in that text as well as a more complete introduction to this subject. The Gelfand-Kirillov dimension (from now on abbreviated as GK-dimension) can be thought of as a measure of the ‘size’ of an algebra, in a similar way as the vector space dimension is used to convey the ‘size’ of a vector space. One may initially object to this, after all $\mathbf{k}$ -algebras are $\mathbf{k}$ -vector spaces so it makes sense to consider the vector space dimension of algebras. However the problem is that even relatively ‘small’ algebras can have infinite vector space dimension. Consider the following example.

Example 12. Let $A = \mathbf{k}[x]$, then the vector space dimension $\dim_{\mathbf{k}}(A) = \infty$ since $x^i \in A$ for all $i \in \mathbb{N}$, and x^i, x^j are linearly independent over $\mathbf{k}$ for $i \neq j$.

Since $\mathbf{k}[x] \subseteq \mathbf{k}[x_1, \dots, x_n]$ it follows that $\dim_{\mathbf{k}}[x_1, \dots, x_n] = \infty$, hence we quickly see that another notion of dimension is needed if we wish to meaningfully use it to distinguish or even classify algebras, in the same vein as vector space dimension is used classify vector spaces up to isomorphism. We proceed to motivate the definition of GK-dimension through a non-rigorous analogy in the hope that it provides, the much sought out after, intuition behind the definition. For those already familiar with the concept, or for those who wish to jump straight to theory please skip the following remark.

Remark 13. Imagine that you are tasked with estimating the (vector space) dimension of Euclidean space $\mathbb{E}^n$ by calculating the lengths/areas/volumes/measures of open balls centred at the origin with radius $r > 0$ in the space. So in the one dimensional case, the open ball is $B_r^1 = \{x \in \mathbb{E}^1 \mid |x| < r\} = (-r, r)$ which is the open interval with radius r , and hence the length is given by $\mu(B_r^1) = 2r$. In $\mathbb{E}^2$, the open ball of radius r is $B_r^2 = \{x \in \mathbb{E}^2 \mid |x| < r\} = \{(x_1, x_2) \in \mathbb{R}^2 \mid x^2 + y^2 < r\}$ which is the open disc with radius r , hence the area is given by $\mu(B_r^2) = \pi r^2$. Analogously, the volume of an open ball is $\lambda(B_r^3) = \frac{4}{3}\pi r^3$. A pattern emerges, and if one were to keep going they would see that for the open n -ball with radius r we have $\mu(B_r^n) = \nu_n r^n$ for some constant

$\nu_n \in \mathbb{R}$. After some thought we spot that the exponent of r in the expressions $\mu(B_r^n)$ matches the dimension of the Euclidean space that the open ball B_r^n inhabits. Taking the logarithm we have, $\ln(\mu(B_r^n)) = n \ln(\nu_n r) = n(\ln(r) + \ln(\nu_n))$, and since ν_n is a constant, by dividing by $\ln(r)$ and taking the limit as $r \rightarrow \infty$ we will cancel out the $\ln(r)$ term in the expression and shrink the $\ln(\nu_n)$ term in the limit. To recap, we have the following

$$\lim_{r \rightarrow \infty} \frac{\ln(\mu(B_r^n))}{\ln(r)} = \lim_{r \rightarrow \infty} \frac{n(\ln(r) + \ln(\nu_n))}{\ln(r)} = n$$

as an expression to estimate the dimension of the Euclidean space $\mathbb{E}^n$ given the open ball B_r^n . By analogously thinking of generating subspaces $V \subseteq A$ of algebras A , we arrive at a similar looking expression in the definition of the GK-dimension below.

Definition 14. The GK-dimension of A is defined as,

$$\text{GKdim}(A) = \sup_V \lim_{n \rightarrow \infty} \frac{\ln(\dim_{\mathbf{k}}(V^n))}{\ln(n)}$$

where the supremum is taken over all generating subspaces $V \subseteq A$.

Remark 15. There exists an even more general definition of GK-dimension of an (left) A -module M , but in this paper we will be content with this less general definition for algebras.

This definition still requires some unpacking to do, first we address the notion of generating subspace.

Definition 16. Let $V \subseteq A$ be a finite subspace of A containing the unit element $1 \in V$, then if V generates A as a $\mathbf{k}$ -algebra, i.e. if every element $f \in A$ may be written as a non-commuting polynomial in $v \in V$ over $\mathbf{k}$, we say that V is a generating subspace of A . We further define $V^n := \mathbf{k} \cdot (v_1 \cdots v_n \mid v_1, \dots, v_n \in V)$.

Next, it can be shown that for affine algebras, which are the ones we are particularly interested in, the dependence on the generating subspace V in the definition of $\text{GKdim}(A)$ can be removed. To see this first let $d_V = \dim_{\mathbf{k}}(V^n)$, and note that $d_V : \mathbb{N} \rightarrow \mathbb{R}$.

Definition 17. Let M_+ denote the set of all functions $f : \mathbb{N} \rightarrow \mathbb{R}$ such that f is eventually monotone increasing and positively valued, that is $\exists N \in \mathbb{N}$ such that $\forall n \geq N$, $f(n) > 0$ and $f(n+1) \geq f(n)$. For $f, g \in M_+$ define $f \preceq g$ if and only if $\exists a, b \in \mathbb{N}$ such that $f(n) \leq ag(bn)$ for almost all $n \in \mathbb{N}$, and define $f \sim g$ if and only if $f \preceq g$ and $g \preceq f$. In fact, $\sim$ forms an equivalence relation on M_+ and for $f \in M_+$ we denote the equivalence class of f as $\mathcal{G}(f) \in M_+ / \sim$. We call $\mathcal{G}(f)$ the growth of f .

Proposition 18. $\sim$ as defined above is an equivalence relation.

Proof. Reflexivity follows since taking $a = b = 1$ we have that $f(n) \leq f(n)$ for all $n \in \mathbb{N}$, hence $f \preceq f$ and $f \sim f$.

Symmetry follows since assuming $f \sim g$ by definition we have $f \preceq g$ and $g \preceq f$, hence again by definition we have that $g \sim f$.

Transitivity follows since assuming $f \sim g$ and $g \sim h$ then we have that $f \preceq g$ and $g \preceq h$ therefore there exists $a_1, b_1, a_2, b_2 \in \mathbb{N}$ such that

$$f(n) \leq a_1 g(b_1 n) \leq a_1 a_2 h(b_2 b_1 n),$$

for almost all $n \in \mathbb{N}$, from which it follows that $f \preceq h$ by taking $a = a_1 a_2$ and $b = b_2 b_1$. Similarly $h \preceq f$, and hence we have $f \sim h$. $\square$

Moreover, we can define a partial order $\prec$ on $M_+ / \sim$ by defining $\mathcal{G}(f) \prec \mathcal{G}(g)$ if and only if $f \preceq g$. The proof of the proposition above shows that $\prec$ is reflexive and transitive, and for antisymmetry we remark that if $\mathcal{G}(f) \prec \mathcal{G}(g)$ and $\mathcal{G}(g) \prec \mathcal{G}(f)$ then by definition $f \sim g$ which is equivalent to $\mathcal{G}(f) = \mathcal{G}(g)$.

Proposition 19. *Let A be affine with generating subspaces V, W , and with d_V, d_W as defined above. Then $\mathcal{G}(d_V) = \mathcal{G}(d_W)$.*

Proof. Since V, W are generating subspaces then by assumption we have

$$\bigcup_{n \in \mathbb{N}} V^n = A = \bigcup_{m \in \mathbb{N}} W^m,$$

hence it follows that for some $a, b \in \mathbb{N}$ we will have $W \subseteq V^a$ and $V \subseteq W^b$. These inclusions imply,

$$W^n \subseteq (V^a)^n = V^{an}, \quad V^n \subseteq (W^b)^n = W^{bn}.$$

It follows that $d_W(n) \leq d_V(an)$ and $d_V(n) \leq d_W(bn)$ for all $n \in \mathbb{N}$, and hence $d_V \sim d_W$, i.e. $\mathcal{G}(d_V) = \mathcal{G}(d_W)$. $\square$

The upshot is that for affine algebras A the choice of generating subspace V is redundant and so we may drop that from the definition of GK-dimension. We now turn our attention to the different types of growth for affine algebras. Let p_m denote the function $p_m(n) = n^m$ and let q_β denote the function $q_\beta(n) = e^{n^\beta}$.

Definition 20. Let A be affine, V a generating subspace of A and define $\mathcal{G}(A) := \mathcal{G}(d_V)$. Note that by Proposition 19 this is well-defined. Then A has,

1. polynomial growth if $\mathcal{G}(A) = \mathcal{G}(p_m)$ for some $m \in \mathbb{N}$,
2. exponential growth if $\mathcal{G}(A) = \mathcal{G}(q_1)$,
3. subexponential growth if $\mathcal{G}(A) \prec \mathcal{G}(q_1)$, and $\nexists m \in \mathbb{N}$ such that $\mathcal{G}(A) \prec \mathcal{G}(p_m)$.

Example 21. Let $A = \mathbf{k}[x]$, and pick $V = \mathbf{k} \cdot (1, x)$ as the generating subspace. We calculate $V^n = \mathbf{k} \cdot (1, x, \dots, x^n)$ hence $d_V(n) = n + 1$, so we see that,

$$\text{GKdim}(A) = \overline{\lim}_{n \rightarrow \infty} \frac{\ln(d_V(n))}{\ln(n)} = \overline{\lim}_{n \rightarrow \infty} \frac{\ln(n+1)}{\ln(n)} = 1.$$

We can generalise the above example to show that $\text{GKdim}(\mathbf{k}[x_1, \dots, x_d]) = d$.

Example 22. Let $A = \mathbf{k}[x_1, \dots, x_d]$, and pick $V = \mathbf{k} \cdot (1, x_1, \dots, x_d)$ as the generating subspace. Then V^n is the subspace spanned by all words in $x_1, \dots, x_d$ of length at most d , some combinatorics relating to counting monomials in commuting variables shows that $d_V(n) = \binom{d+n}{d}$ which we note is a polynomial in n of degree d . More precisely, we have

$$\binom{d+n}{d} = \frac{(d+n)!}{n!d!} = \frac{(n+d) \cdots (n+1)}{d!} = \frac{n^d + \cdots + d!}{d!},$$

and after some thought we may come up with the expressions $n^d/d!$ and $(d+1)n^d$ which we claim are lower and upper bounds respectively. For the lower bound we wish to show that

$$\frac{n^d}{d!} \leq \frac{(n+d) \cdots (n+1)}{d!},$$

which is equivalent to $n^d \leq (n+d) \cdots (n+1)$. This inequality is true since both sides of the inequality are d -fold products and we can pair the factors either side of the inequality such that $n \leq n+i$ for $1 \leq i \leq d$, hence it follows that the product on the right is at least the product on the left. Therefore $n^d/d!$ is a lower bound. Similarly for the upper bound we wish to show that,

$$(d+1)n^d \geq \frac{(n+d) \cdots (n+1)}{d!} \tag{2.1}$$

which is equivalent to $(d+1)n^d \geq (n+d) \cdots (n+1)$. Again, we have n -fold products either side of the inequality and we pair the factor $(i+1)n$ together with the factor $(n+i)$, so the target inequality holds if for $1 \leq i \leq d$ we have that $(i+1)n \geq n+i$ holds, which it clearly does. Therefore $(d+1)n^d$ is an upper bound as claimed.

Since $\ln$ is an increasing function the bounds above imply that

$$\frac{\ln(n^d/d!)}{\ln(n)} \leq \frac{\ln(d_V(n))}{\ln(n)} \leq \frac{\ln((d+1)n^d)}{\ln(n)},$$

and note

$$\overline{\lim}_{n \rightarrow \infty} \frac{\ln(n^d/d!)}{\ln(n)} = d, \quad \overline{\lim}_{n \rightarrow \infty} \frac{\ln((d+1)n^d)}{\ln(n)} = d,$$

hence by the Squeeze Theorem it follows that $\text{GKdim}(A) = d$.

So far the examples we have considered include algebras with polynomial growth, we now consider algebras with other types of growth.

Example 23. Let $A = \mathbf{k}\langle x, y \rangle$, and pick $V = \mathbf{k} \cdot (1, x, y)$ as the generating subspace. Then V^n is the subspace spanned by all words in the **non-commuting** variables x, y of length at most n . We can relate words of length n to n tuples where for each entry we have two choices (x or y), there are exactly 2^n such tuples for each n , and hence by summing up to n we see that

$$d_V(n) = \sum_{i=0}^n 2^i = (1 + 1 + 2 + \cdots + 2^n) - 1 = 2^{n+1} - 1 = e^{\ln(2)(n+1)} - 1.$$

Recall that we use the notation $q_1(n) = e^n$. Note that

$$d_V(n) = e^{\ln(2)(n+1)} - 1 < e^{\ln(2)(n+1)} \leq e^{2n} = q_1(2n)$$

for almost all n , where the second inequality follows since the exponential is an increasing function and $\ln(2)(n+1) \leq 2n$ for almost all n . It follows that $d_V \preceq q_1$. Additionally, we claim that $q_1(n) \leq d_V(2n)$ for almost all n . Observe that

$$d_V(2n) - q_1(n) = e^{\ln(2)(2n+1)} - 1 - e^n,$$

and note that for $n = 0$ the above is equal to $e^{\ln(2)} - 1 - 1 = 0 \geq 0$. Moreover, the derivative of $d_V(2n) - q_1(n)$ with respect to n is given by

$$2 \ln(2) e^{\ln(2)(2n+1)} - e^n > e^{\ln(2)(2n+1)} - e^n \geq 0,$$

where the second inequality follows since the exponential is an increasing function and $\ln(2)(2n+1) \geq n$ for all $n \in \mathbb{N}$. Since $d_V(2n) - q_1(n)$ is continuous as a difference of continuous functions, it follows from the above that $d_V(2n) - q_1(n) \geq 0$ for all $n \in \mathbb{N}$, thus $q_1(n) \leq d_V(2n)$ for all $n \in \mathbb{N}$. It follows that $d_V \preceq q_1$. It now follows that $\mathcal{G}(A) = \mathcal{G}(d_V) = \mathcal{G}(q_1)$ and hence A has exponential growth. Moreover,

$$\text{GKdim}(A) = \overline{\lim}_{n \rightarrow \infty} \frac{\ln(2^{n+1} - 1)}{\ln(n)} = \overline{\lim}_{n \rightarrow \infty} \frac{(n+1) \ln(2)}{\ln(n)} = \infty,$$

Remark 24. It is a good time to point out that any affine algebra with exponential growth will have infinite GK-dimension. This follows since,

$$\overline{\lim}_{n \rightarrow \infty} \frac{\ln(q_1)}{\ln(n)} = \overline{\lim}_{n \rightarrow \infty} \frac{\ln(e^n)}{\ln(n)} = \overline{\lim}_{n \rightarrow \infty} \frac{n}{\ln(n)} = \infty.$$

Example 25. Let $\mathfrak{g}$ be the Lie algebra with infinite basis $B = \{x, y_1, y_2, \dots\}$ and relations $[x, y_i] = y_{i+1}, [y_i, y_j] = 0$ for all $i, j \geq 1$. Take the universal enveloping algebra

$U(\mathfrak{g})$ of $\mathfrak{g}$, which is defined as

$$U(\mathfrak{g}) = \frac{\mathbf{k}\langle x, y_1, y_2, \dots \rangle}{(st - ts - [s, t] \mid s, t \in B)}.$$

Then $A = U(\mathfrak{g})$ has generating subspace $V = \mathbf{k} \cdot (1, x, y_1)$ and it can be shown that $\mathcal{G}(d_V(n)) = \mathcal{G}(e^{\sqrt{n}})$ [see [KL00] Section 12.1], from which it follows that A has subexponential growth, and infinite GK-dimension.

Remark 26. It is also a good time to point out that any affine algebra with subexponential growth will have infinite GK-dimension. To see why, assume for a contradiction that A is an affine algebra with subexponential growth and $\text{GKdim}(A) = r$ for some finite $r > 0$. Then take $m = \lceil r \rceil \in \mathbb{N}$ and note that

$$\lim_{n \rightarrow \infty} \frac{\ln(n^m)}{\ln(n)} = m \geq r = \lim_{n \rightarrow \infty} \frac{\ln(d_V(n))}{\ln(n)} = \text{GKdim}(A),$$

which implies that $\mathcal{G}(d_V) \leq \mathcal{G}(p_m)$, but this contradicts our assumption that A has subexponential growth.

Chapter 3

The Stephenson-Zhang Counterexample

Here we go over the theory developed by Stephenson & Zhang which they used to present their counterexample to the MR question. All results in this chapter come from [SZ97], and the aim of this chapter will be to present their results and expand upon their proofs where appropriate. We will first prove a technical lemma needed to prove a theorem about the growth of graded, locally finite, right Noetherian rings. As a corollary of this theorem we will see that affine, right Noetherian algebras with exponential growth do not admit finite dimensional Noetherian filtrations, and we will give an example of such a ring which was the counterexample to the MR question given by Stephenson and Zhang.

Lemma 27 ([SZ97] Lemma 1.1). *Let $\{d_n \mid n \in \mathbb{N}\}$ be a sequence of non-negative integers with infinitely many non-zero d_n 's. Then,*

1. $\overline{\lim}_{n \rightarrow \infty} (\sum_{i \leq n} d_i)^{1/n} = \overline{\lim}_{n \rightarrow \infty} d_n^{1/n}$
2. Suppose $d_n > 0$ for all $n \in \mathbb{N}$. Given $l_0, l_1, \dots, l_s \in \mathbb{Z}_+$,

$$\overline{\lim}_{n \rightarrow \infty} \min\{(d_n/d_{n-l_j})^{1/l_j} \mid j = 0, 1, \dots, s\} \geq \overline{\lim}_{n \rightarrow \infty} d_n^{1/n}$$

3. If $\overline{\lim}_{n \rightarrow \infty} d_n^{1/n} > 1$, then there exists a subsequence $\{l_i \mid i \in \mathbb{N}\} \subset \mathbb{N}$ such that $d_{l_0} > 1$ and $d_{l_s} > \sum_{i < s} d_{l_s - l_i}$ for all $s \geq 1$.

Proof. For (1), first note that since $\sum_{i \leq n} d_i \geq d_n$ we immediately have,

$$\overline{\lim}_{n \rightarrow \infty} \left(\sum_{i \leq n} d_i \right)^{1/n} \geq \overline{\lim}_{n \rightarrow \infty} d_n^{1/n}$$

Let $L = \overline{\lim}_{n \rightarrow \infty} d_n^{1/n}$, if $L = \infty$ then the above inequality implies the result. Hence let us assume that $L < \infty$. Since the sequence $\{d_n\}_{n \in \mathbb{N}}$ has infinitely many non-zero elements,

i.e. $\forall n \in \mathbb{N}, \exists m \geq n$ such that $d_m \geq 1$, it follows that $\forall n \in \mathbb{N}, \sup_{m \geq n} \{d_m^{1/m}\} \geq 1$. Therefore $\overline{\lim}_{n \rightarrow \infty} d_n^{1/n} \geq \overline{\lim}_{n \rightarrow \infty} 1^{1/n} \geq 1$, hence $L \geq 1$. Now suppose that $L < \alpha$ for some $\alpha > 1$, then $\exists n_0 \in \mathbb{N}$ such that $d_n < \alpha^n$ for all $n \geq n_0$, and hence we can split the sum $\sum_{i \leq n} d_i$ as follows

$$\sum_{i \leq n} d_i = \sum_{i \leq n_0} d_i + \sum_{i=n_0+1}^n d_i < \left(\sum_{i \leq n_0} d_i \right) + n\alpha^n.$$

We thus obtain the following inequality:

$$\begin{aligned} \overline{\lim}_{n \rightarrow \infty} \left(\sum_{i \leq n} d_i \right)^{1/n} &\leq \overline{\lim}_{n \rightarrow \infty} \left(\left(\sum_{i \leq n_0} d_i \right) + n\alpha^n \right)^{1/n} \\ &\leq \overline{\lim}_{n \rightarrow \infty} \left(\left(\sum_{i \leq n_0} d_i \right)^{1/n} + (n\alpha^n)^{1/n} \right) \\ &\leq \overline{\lim}_{n \rightarrow \infty} (n\alpha^n)^{1/n} \\ &= \alpha \overline{\lim}_{n \rightarrow \infty} n^{1/n} \\ &= \alpha, \end{aligned}$$

where the third inequality follows since $\left(\sum_{i \leq n_0} d_i \right)$ is finite and hence it will disappear in the limit and we note that $\overline{\lim}_{n \rightarrow \infty} n^{1/n} = 1$ follows since,

$$\ln \left(\lim_{n \rightarrow \infty} n^{1/n} \right) = \lim_{n \rightarrow \infty} \ln \left(n^{1/n} \right) = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \ln(n) \right) = 0,$$

which implies that $\lim_{n \rightarrow \infty} n^{1/n} = e^0 = 1$ and hence $\overline{\lim}_{n \rightarrow \infty} n^{1/n} = 1$. This holds for arbitrary $\alpha > 1$, thus taking the limit from the right as $\alpha \rightarrow 1$ we conclude that $\overline{\lim}_{n \rightarrow \infty} \left(\sum_{i \leq n} d_i \right)^{1/n} \leq 1$, and combining this with inequality $L \geq 1$ we obtain

$$\overline{\lim}_{n \rightarrow \infty} \left(\sum_{i \leq n} d_i \right)^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} d_n^{1/n}.$$

We have now shown both inequalities, hence the result follows.

For (2), we suppose in addition to our previous assumptions for (1) that $d_n > 0$ for all $n \in \mathbb{N}$. Let $L' = \overline{\lim}_{n \rightarrow \infty} \min\{(d_n/d_{n-l_j})^{1/l_j} \mid j = 0, 1, \dots, s\}$, if $L' = \infty$ then the result follows trivially, hence we assume that $L' < \infty$. The result we wish to show, $L' \geq L$, is equivalent to showing that if for some $\alpha > 1$, we have $L' < \alpha$ then $L \leq \alpha$. Let us assume that $L' < \alpha$ for $\alpha > 1$, it suffices to prove that $L \leq \alpha$. Our assumption

implies that $\exists m_0 \in \mathbb{N}$ such that $\min\{(d_n/d_{n-l_j})^{1/l_j} \mid j = 0, 1, \dots, s\} < \alpha$ for all $n \geq m_0$. Let $i \in \{0, 1, \dots, s\}$ be the index at which the minimum is achieved, manipulating $(d_n/d_{n-l_i})^{1/l_i} < \alpha$ we see that

$$d_n < \alpha^{l_i} d_{n-l_i}, \quad \forall n \geq m_0. \quad (3.1)$$

Now we make the following claim:

$$d_n < \max_{0 \leq j \leq m_0} \{\alpha^{n-j} d_j\}, \quad \forall n \geq m_0.$$

which we prove by induction on n .

Proof (of claim). For the base case $n = m_0$ we can use Inequality 3.1 to see that $d_{m_0} < \alpha^{l_i} d_{m_0-l_i}$, where we observe that $0 \leq m_0 - l_i \leq m_0$ since we can only have non-negative indices of the sequence (which gives the lower bound) and since $l_i \geq 1$ by assumption (which gives the upper bound). Setting $j = m_0 - l_i$, we first see that $j \in \{0, 1, \dots, m_0\}$ as well as

$$d_{m_0} < \alpha^{l_i} d_{m_0-l_i} = \alpha^{m_0-j} d_j \leq \max_{0 \leq j \leq m_0} \{\alpha^{m_0-j} d_j\},$$

which completes the base case. For the inductive step, we assume the inductive hypothesis $d_k < \max_{0 \leq j \leq m_0} \{\alpha^{k-j} d_j\}$ for $m_0 \leq k \leq n$, and consider the case $k = n + 1$. We can again use Inequality 3.1 to obtain $d_{n+1} < \alpha^{l_i} d_{n+1-l_i}$, and since by assumption $l_i \geq 1$ we note that $n + 1 - l_i \leq n$, hence we can apply the inductive hypothesis to d_{n+1-l_i} to obtain the following chain of inequalities

$$d_{n+1} < \alpha^{l_i} d_{n+1-l_i} < \alpha^{l_i} \max_{0 \leq j \leq m_0} \{\alpha^{n+1-l_i-j} d_j\} = \max_{0 \leq j \leq m_0} \{\alpha^{n+1-j} d_j\}.$$

This proves the claim for the case $k = n + 1$, and so completes the inductive step. It now follows by induction that the claim holds $\forall n \geq m_0$. $\square$

Let β denote the maximum value of j for which we achieve the maximum in the claim we have just proven, i.e. let $\alpha^{n-\beta} d_\beta = \max_{0 \leq j \leq m_0} \{\alpha^{n-j} d_j\}$. Then $\forall n \geq m_0$ we have $d_n < d_\beta \alpha^{n-\beta}$ which implies $d_n^{1/n} < (d_\beta)^{1/n} (\alpha^{1-\beta/n})$. Since $d_\beta \in \mathbb{N}$ is a constant we have $d_\beta < n$ eventually for all big n , hence we can further bound $d_n^{1/n}$ above by $d_n^{1/n} < d_\beta^{1/n} (\alpha^{1-\beta/n}) < n^{1/n} (\alpha^{1-\beta/n})$. The existence of the limits

$$\lim_{n \rightarrow \infty} n^{1/n} = 1, \quad \text{and} \quad \lim_{n \rightarrow \infty} \alpha^{1-\beta/n} = \alpha,$$

implies the existence of the limit suprema which are equal to their respective limits above. Hence putting everything together we obtain the result,

$$L = \overline{\lim}_{n \rightarrow \infty} d_n^{1/n} \leq \overline{\lim}_{n \rightarrow \infty} n^{1/n} \alpha^{1-\beta/n} = \overline{\lim}_{n \rightarrow \infty} n^{1/n} \cdot \overline{\lim}_{n \rightarrow \infty} \alpha^{1-\beta/n} = 1 \cdot \alpha = \alpha.$$

By our earlier remarks (2) now follows.

For (3), suppose $\overline{\lim}_{n \rightarrow \infty} d_n^{1/n} =: a > 1$ and pick some b such that $1 < b < a$ (we could pick $b = \frac{a+1}{2}$ for example). Therefore every possible tail of the sequence has an element strictly greater 1, hence let $l_0 \in \mathbb{N}$ such that $d_{l_0} > 1$ and $b^{l_0} > 2$. We proceed to construct the sequence $\{l_i\}_{i \in \mathbb{N}}$ inductively in the following way. Suppose we have obtained all elements of the sequence up to and including l_s , that is suppose we have positive integers $l_0 < l_1 < \dots < l_s$ such that $d_{l_j} > \sum_{i < j} d_{l_j - l_i}$ and $b^{l_j} > 2^{j+1}$ for all $j \geq 1$. Since the subsequence we are looking for satisfies $d_{l_s} > 1$ for all s , it is valid to consider the alternative sequence (which for ease of notation we will also denote d_n) for which every 0 is replaced with 1 such that $d_n > 0$ for all n . The idea being that the same indices in the original sequence will give us the desired subsequence. Now, this new sequence satisfies the hypothesis of (2) of this lemma, hence we have that

$$\overline{\lim}_{n \rightarrow \infty} \min\{(d_n/d_{n-l_j})^{1/l_j} \mid j = 0, 1, \dots, s\} \geq a > b,$$

therefore there exists a subsequence $\{n_i\}_{i \in \mathbb{N}}$ such that for each term of this subsequence

$$\min\{(d_{n_i}/d_{n_i-l_j})^{1/l_j} \mid j = 0, 1, \dots, s\} > b. \quad (3.2)$$

Now let l_{s+1} be an element of the subsequence $\{n_i\}_{i \in \mathbb{N}}$ so that $l_{s+1} > l_s$ and $b^{l_{s+1}} > 2^{s+2}$. Using Inequality 3.2 and using the inductive hypothesis on the sequence terms with index strictly less than $s+1$, we obtain the following chain of inequalities

$$\sum_{i < s+1} d_{l_{s+1}-l_i} < \sum_{i < s+1} d_{l_{s+1}} b^{-l_i} = d_{l_{s+1}} \sum_{i < s+1} b^{-l_i} < d_{l_{s+1}} \sum_{i < s+1} 2^{-(i+1)} < d_{l_{s+1}} \sum_{i=0}^{\infty} 2^{-(i+1)},$$

and noting that $\sum_{i=0}^{\infty} 2^{-(i+1)} = 1$ the above implies that $\sum_{i < s+1} d_{l_{s+1}-l_i} < d_{l_{s+1}}$. Hence by induction the desired sequence exists and we are done. $\square$

Before proceeding to the main theorem, we require the following definition.

Definition 28. Let $A = \bigoplus_{i=0}^{\infty} A_i$ be a graded algebra, then A is locally finite if $\dim_{\mathbf{k}}(A_i) < \infty$ for all $i \in \mathbb{N}$.

With this technical lemma under our belts, we proceed to prove the following powerful theorem.

Theorem 29. *[[SZ97] Theorem 1.2] If A is a graded locally finite right Noetherian $\mathbf{k}$ -algebra, then A has subexponential growth.*

Proof. Let $d_n := \dim_{\mathbf{k}}(A_n)$. For a contradiction we will assume that A has exponential growth, i.e. that $\overline{\lim}_{n \rightarrow \infty} (\sum_{i \leq n} d_i)^{1/n} =: L > 1$. By part 1 of Lemma 27 we have that $\overline{\lim}_{n \rightarrow \infty} d_n^{1/n} = L > 1$, and hence we can apply part 3 of the above Lemma to obtain a subsequence $\{l_i\}_{i \in \mathbb{N}}$ such that $d_{l_0} > 1$ and $d_{l_s} > \sum_{i < s} d_{l_s - l_i}$ for all $s \geq 1$. Note that for this subsequence it follows that $d_{l_i} > 1$. We now inductively define a sequence of right

ideals, let $I^s := \sum_{i \leq s} x_i A$ for some $x_i \in A_{l_i}$. First note that $I^0 = x_0 A$ where $x_0 \in A_{l_0}$, and we can pick an element $x_0 \in A_{l_0}$ since $\dim_{\mathbf{k}}(A_{l_0}) = d_{l_0} > 1$ and hence it is non-empty. Now suppose that we have gotten $x_{l_i} \in A_{l_i}$ for all $i < s$. Since $d_{l_s} > \sum_{i < s} d_{l_s - l_i}$ we obtain the following chain of inequalities

$$\dim_{\mathbf{k}}(A_{l_s}) > \sum_{i < s} \dim_{\mathbf{k}}(A_{l_s - l_i}) \geq \sum_{i < s} \dim_{\mathbf{k}}((x_i A)_{l_s}) \geq \dim_{\mathbf{k}}((I^{s-1})_{l_s}),$$

where we note that the second inequality follows as we have a surjection $A_{l_s - l_i} \rightarrow (x_i A)_{l_s}$ of $\mathbf{k}$ -subspaces given by $a \mapsto x_i a$ which implies $\dim_{\mathbf{k}}(A_{l_s - l_i}) \geq \dim_{\mathbf{k}}((x_i A)_{l_s})$ for all $i < s$. Moreover, the third inequality above follows as we have a clear surjection $\bigoplus_{i < s} x_i A \rightarrow \sum_{i < s} x_i A = I^{s-1}$. This strict dimensional inequality justifies us in picking an element $x_s \in A_{l_s} \setminus I^{s-1}$, which we use to form the ideal $I^s = x_s A + I^{s-1}$. Note that $I^{s-1} \subset I^s$ is strict. Thus we have inductively obtained a strictly increasing chain of right ideals of A which does not terminate, $\{I^s\}_{s \in \mathbb{N}}$, which clearly violates our assumption that A is right Noetherian. By contradiction it now follows that A has subexponential growth. $\square$

A corollary of this theorem is the following partial answer to the MR question.

Corollary 30 ([SZ97] Corollary 1.3). *Let R be a finitely generated right Noetherian algebra with exponential growth and let $F_{\bullet} := \{F_n \mid n \in \mathbb{N}\}$ be a finite dimensional filtration of R . Then the associated graded ring $\text{gr}_F R$ is neither left nor right Noetherian.*

Proof. Since R is finitely generated it follows that F_m (a finite dimensional subspace which contains the identity 1) generates R as an algebra, for some $m \in \mathbb{N}$. Since R has exponential growth this means by the definition that

$$\overline{\lim}_{n \rightarrow \infty} (\dim_{\mathbf{k}}((F_m)^n))^{1/n} > 1. \quad (3.3)$$

Note that since $F_{\bullet}$ is a filtration we have $F_m^n \subseteq F_{\sum_{i=1}^n m} = F_{nm}$. Let us denote $L = \overline{\lim}_{n \rightarrow \infty} (\dim_{\mathbf{k}}(F_n))^{1/n}$, we will want to show that $L > 1$ so for a contradiction let us assume that $L \leq 1$. This implies that $\forall n \in \mathbb{N}$ we have $\sup_{N \geq n} (\dim_{\mathbf{k}}(F_N))^{1/N} \leq 1$ which implies in turn that $\sup_{N \geq n} \dim_{\mathbf{k}}(F_N) \leq 1$. Therefore $\dim_{\mathbf{k}}(F_n) \leq 1$ for all $n \in \mathbb{N}$. Furthermore, since $F_{\bullet}$ is a filtration it follows that $\mathbf{k} \subseteq F_0$ and $F_n \subseteq F_{n+1}$ for all $n \in \mathbb{N}$, hence $\dim_{\mathbf{k}}(F_0) \geq 1$ and $\dim_{\mathbf{k}}(F_{n+1}) \geq \dim_{\mathbf{k}}(F_n) \geq 1$. Altogether we deduce that $\dim_{\mathbf{k}}(F_n) = 1$ for all $n \in \mathbb{N}$, which implies that $\dim_{\mathbf{k}}((F_m)^n) = 1$, but this quickly contradicts Inequality 3.3. It follows that $L > 1$.

By definition, $\text{gr}_F R = \bigoplus_{i \in \mathbb{N}} (F_i / F_{i-1})$, hence using the fact that this is a graded ring we see that for R to have exponential growth the limit

$$\overline{\lim}_{n \rightarrow \infty} \left(\sum_{i \leq n} \dim_{\mathbf{k}}(F_i / F_{i-1}) \right)^{1/n}$$

must be strictly greater than one. Note that the sum in the limit above is a telescoping sum,

$$\begin{aligned} \sum_{i \leq n} \dim_{\mathbf{k}}(F_i/F_{i-1}) &= \sum_{i \leq n} \dim_{\mathbf{k}}(F_i) - \dim_{\mathbf{k}}(F_{i-1}) \\ &= \dim_{\mathbf{k}}(F_n) - \dim_{\mathbf{k}}(F_{n-1}) + \cdots + \dim_{\mathbf{k}}(F_1) - \dim_{\mathbf{k}}(F_0), \end{aligned}$$

using which we see that the limit above reduces to,

$$\overline{\lim}_{n \rightarrow \infty} (\dim_{\mathbf{k}}(F_n) - \dim_{\mathbf{k}}(F_0))^{1/n}.$$

Since $\dim_{\mathbf{k}}(F_0) \in \mathbb{N}$ is some constant (remember the assumption that we are working with a finite dimensional filtration) we see that this limit reduces to,

$$\overline{\lim}_{n \rightarrow \infty} (\dim_{\mathbf{k}}(F_n))^{1/n} = L > 1.$$

It follows that $\text{gr}_F R$ has exponential growth, and so by the contrapositive of Theorem 29 and noting that by construction $\text{gr}_F R$ is graded and locally finite it follows that $\text{gr}_F R$ is not Noetherian. $\square$

We are now ready to present the Stephenson-Zhang counterexample.

Example 31. [[SZ97] Example 1.4] Consider the following skew polynomial ring,

$$R := \mathbf{k}[x, y][z, z^{-1}, \sigma] = \frac{\mathbf{k}[x, y][z, z^{-1}]}{(fz - zf, fz^{-1} - z^{-1}f \mid f \in \mathbf{k}[x, y])}$$

where $\sigma : \mathbf{k}[x, y] \rightarrow \mathbf{k}[x, y]$ is an automorphism of $\mathbf{k}[x, y]$ given by $\sigma(x) = y + x^2$ and $\sigma(y) = x$. Since $\mathbf{k}[x, y]$ is Noetherian it follows that R is Noetherian, we will cover this implication in more detail in Section 4.1. It is also immediate that R is finitely generated by x, y, z, z^{-1} . Smith has shown in 1982 [cf. [Smi82] Lemma 10] that R has exponential growth. We now see that Corollary 30 implies that R does not admit a finite dimensional Noetherian filtration $F_{\bullet}$.

Therefore R is an example of an affine Noetherian algebra which does not admit a finite dimensional Noetherian filtration, which provides a clear counterexample to the MR question.

Chapter 4

The MR Question and the MS Example

In this chapter we will introduce the $\mathbf{k}$ -algebra S which was considered by McConnell and Stafford in [MS89], and present some results about this algebra which will allow us to comment on the applicability of the results mentioned in the introduction to S . Moreover, we will define the special class of filtrations which we will refer to as McConnell-Stafford filtrations (or MS filtrations for short), prove some properties of this class of filtrations, and in so doing set up the ground work for the main results of this paper to be presented in Chapter 5. Note that we will sometimes refer to S as the MS example.

We now go on to give a brief overview of the partial results obtained by various authors, and introduce the algebra S for which it is not known whether all finite dimensional filtrations are not Noetherian.

4.1 The MS Example: S

First let us define S . Let $R = \mathbf{k}[y, z, z^{-1}]$ thought of as a $\mathbf{k}$ -algebra, and let $\sigma \in \text{Aut}(R)$ (note that σ is $\mathbf{k}$ -linear) given by $\sigma(z) = z$ and $\sigma(y) = zy$. Then we can define $S = R[x; \sigma]$ as an Ore extension of R . We relabel $w = z^{-1}$, so that a presentation of S is given by,

$$S = \frac{\mathbf{k}\langle x, y, z, w \rangle}{(xyz - yx, zw - 1, \text{ and } z, w \text{ central})}$$

We now present a series of facts without proof, all of which can be found in [MR01], which we will use to show some relevant properties of S we will wish to discuss.

Theorem 32. *Let R_1, R_2 be rings such that $R_2 = R_1[x; \sigma, \delta]$ is a skew polynomial ring. Then the following hold,*

i If σ is injective and R_1 is a domain, then R_2 is a integral domain.

- ii If σ is an automorphism and R_1 is a right (or left) Noetherian ring, then R_2 is a right (respectively left) Noetherian ring.

Proof. See [MR01] Theorem 1.2.9 for proof. $\square$

Theorem 33. *Let R be a Noetherian domain, then R is Goldie.*

Proof. See [MR01] Theorem 2.1.15, Lemma 2.2.5, and Definition 2.3.1. $\square$

Remark 34. For the sake of brevity we will not rigorously define what it means for a ring R to be Goldie, for a rigorous treatment of this we direct the reader to chapter 2 of [MR01]. For now, one can think of a ring R being right Goldie if R as a right R -module does not contain an infinite direct sum of submodules and if R satisfies the ascending chain condition on annihilator right ideals. Appropriately replacing “right” with “left” gives the notion of a left Goldie ring, and a ring is Goldie if it is both left and right Goldie.

We also recall the definition of a prime ring,

Definition 35 (cf. [MR01] Proposition 0.2.3 (iii)). Let R be a ring, then R is a prime ring if 0 is a prime ideal.

We make the following observations about S .

1. S is Noetherian by Theorem 32 (ii) since $S = R[x; \sigma, \delta]$, R is Noetherian and σ is an automorphism.
2. $\text{GKdim}(S) = 4$, by [MR01] Example 8.2.16. Note that finite GK dimension implies that S has subexponential growth.
3. S is Goldie. To see this note that by Theorem 32 (i), since R is a domain and σ is injective, it follows that S is a domain. It now follows that S is a Noetherian domain and hence by Theorem 33 it follows that S is Goldie.
4. S is a prime ring since S is a domain, which follows from Definition 35.

We will want to talk about minimal generating sets of algebras, and to avoid confusion over the use of the word “minimal” we make the following definition.

Definition 36. Let A be an algebra and let $g_1, \dots, g_n \in A$ such that they generate A as an algebra. Then we say that $G = \{g_1, \dots, g_n\}$ is a minimal generating set of A if $\nexists g'_1, \dots, g'_m \in A$ such that $g'_1, \dots, g'_m$ generate A as an algebra and $m < n$. If $G = \{g_1, \dots, g_n\}$ is a minimal generating set of A we might alternatively say that $g_1, \dots, g_n$ minimally generate A .

Remark 37. Whilst it is clear from the presentation above that S is finitely generated in four elements, we do not know if it is minimally generated in four elements. That is, we do not know whether there exist any three elements of S which generate it as an algebra. This seems like a **hard** problem, and going forward we will assume that S is minimally generated in four elements, however full-disclosure that this is an assumption!

4.2 Partial Results for the MR Question

Having introduced the MS example, we now present the statements, without proof, of the results which give partial answers to the MR question.

4.2.1 Growth of Graded Noetherian Rings

We have already seen the key results with regard to the MR question, namely Theorem 1.2 and Corollary 1.3, of this paper in Chapter 3. As remarked earlier, S is an affine right Noetherian algebra with **subexponential growth**, hence Corollary 1.3 does not apply to S .

4.2.2 Algebras without Noetherian Filtrations

The main results we wish to focus on from this paper include the following,

Theorem 38 ([SZ03] Theorem 1.1). *Suppose that I , and $J_1 \subset J_2$ are ideals of an algebra R such that J_2/J_1 is free of rank s as a left R/I -module and free of rank t as a right R/I -module. If $s < t$, then there is no finite $\mathbb{N}$ -filtration Γ of R such that $\text{gr}_\Gamma S$ is left Noetherian.*

Theorem 39 ([SZ03] Theorem 3.2). *Suppose that I , and $J_1 \subset J_2$ are ideals of an algebra R such that $A = R/I$ is a prime Goldie ring. Assume that J_2/J_1 has Goldie rank s as a left A -module Goldie rank t as a right A -module, for some $s < t$. Then there is no finite $\mathbb{N}$ -filtration Γ' of R such that $\text{gr}_{\Gamma'} S$ is left Noetherian.*

We recall that from our previous remarks we have concluded that S is a prime Goldie ring, so taking $I = 0$ we would have $A = S/I = S/0 \cong S$ in the above theorem. Now, if we could find ideals $J_1 \subset J_2$ which satisfy the hypothesis of Theorem 39 we would have an answer to the MR question for S . Whilst a promising sign, we were not able to find such ideals J_1, J_2 and we were not able to show that we cannot find appropriate ideals I, J_1, J_2 of S which satisfy the hypothesis of Theorem 39, therefore we are unsure whether we can apply Theorem 39 (and for similar reasons Theorem 38) to S .

4.2.3 Affine Noetherian Algebras, Filtrations and Presentations

The main results we wish to focus on from this paper include the following.

Theorem 40 ([Gre23] Theorem 1.1). *Let R be an algebra and let F be a finite-dimensional filtration. If $\text{gr}_F S$ is Noetherian then R is finitely presented.*

If S was not finitely presented, then we could use Theorem 40 to prove by contradiction that S has no finite-dimensional Noetherian filtrations. However as remarked earlier, S is finitely presented and thus we can not apply Theorem 40 to settle the MR question for S .

4.3 MS Filtrations

The material in this section builds upon the material on filtrations and gradings introduced in Chapter 1, we encourage the reader to review this earlier material before proceeding with this section.

4.3.1 Preliminaries

We recall an important definition regarding morphisms of filtered algebras, a good source for this is Section 7.6 of [MR01].

Definition 41 (cf. [MR01] 7.6.12). Let A and B be $\mathbf{k}$ -algebras with filtrations $F_\bullet$ and $F'_\bullet$ respectively. Let $\psi : A \rightarrow B$ be an algebra morphism, then ψ is a filtered morphism if $\psi(F_n) \subseteq F'_n \cap \psi(A)$ for all $n \in \mathbb{N}$. Moreover, we say that ψ is strict if $\psi(F_n) = F'_n \cap \psi(A)$. Note in particular that filtered and surjective morphisms are strict.

Now note that for any filtered algebra A with filtration $F_\bullet$ we have that $A = \bigcup_{n \in \mathbb{N}} F_n$, hence $\forall h \in A$, $h \in F_m$ for some $m \in \mathbb{N}$ so it follows that the set $\{n \in \mathbb{N} \mid h \in F_n\}$ is non-empty. Since $\mathbb{N}$ is a well-ordered set, we may ask for the minimal n such that $h \in F_n$. This motivates the following definition, which will also serve as a notational shorthand.

Definition 42. Let A be a filtered algebra with filtration $F_\bullet$, for $h \in A$ we define $\deg_F(h) := \min\{n \in \mathbb{N} \mid h \in F_n\}$. We call $\deg_F(h)$ the “ F -degree of h ”.

Given two filtered algebras A, B with filtrations $F_\bullet, F'_\bullet$ respectively, and a filtered morphism $\psi : A \rightarrow B$, one may wonder if there exists a morphism between $\text{gr}_F A \rightarrow \text{gr}_{F'} B$. Remarkably this turns out to be the case, that is given the set up above there exists an associated algebra morphism which we denote $\text{gr}\psi : \text{gr}_F A \rightarrow \text{gr}_{F'} B$. We show this in the following construction.

Construction 43. Let A, B be filtered algebras with filtrations $F_\bullet, F'_\bullet$ and a filtered morphism $\psi : A \rightarrow B$. To begin we concern ourselves with the case where for some $N \in \mathbb{N}$, $h \in F_N/F_{N-1}$ is homogenous. Since $\text{can}_{F_{N-1}}$ is surjective, we can choose some preimage $p \in \text{can}_{F_{N-1}}^{-1}(h)$, hence using the assumption that ψ is filtered we have $\psi(p) \in F'_N$. We can now map $\psi(p)$ using $\text{can}_{F'_{N-1}}$ to $\text{can}_{F'_{N-1}}(\psi(p)) \in F'_N/F'_{N-1}$, and in this way we can define a map $(\text{gr}\psi)_N : F_N/F_{N-1} \rightarrow F'_N/F'_{N-1}$ which maps h to $\text{can}_{F'_{N-1}}(\psi(p))$ where we have made a choice of $p \in \text{can}_{F_{N-1}}^{-1}(h)$. Of course, having made a choice of the preimage of h we have to worry whether the map we have just defined is well-defined. So let $p_1, p_2 \in \text{can}_{F_{N-1}}^{-1}(h)$, we want to show that $\text{can}_{F'_{N-1}}(\psi(p_1)) = \text{can}_{F'_{N-1}}(\psi(p_2))$. Taking the difference and remarking that $\text{can}_{F_{N-1}} \circ \psi$ is a $\mathbf{k}$ -linear map as a composition of $\mathbf{k}$ -linear maps, we note that this is equivalent to showing, $\text{can}_{F'_{N-1}}(\psi(p_1 - p_2)) = 0$. By definition of the canonical surjection $\text{can}_{F'_{N-1}}$, it suffices to show $\psi(p_1 - p_2) \in F'_N$ and since ψ is filtered it suffices to show that $p_1 - p_2 \in F_{N-1}$. By assumption p_1, p_2 are the

preimages of h under $\text{can}_{F_{N-1}}$, hence using the fact that this is also a $\mathbf{k}$ -linear map we see that

$$\text{can}_{F_{N-1}}(p_1 - p_2) = \text{can}_{F_{N-1}}(p_1) - \text{can}_{F_{N-1}}(p_2) = h - h = 0.$$

Since $\ker(\text{can}_{F_{N-1}}) = F_{N-1}$, it follows that $p_1 - p_2 \in F_{N-1}$. Thus $(\text{gr}\psi)_N$ is well-defined. But more is true, this map is in fact also $\mathbf{k}$ -linear which we show now. Let $h_1, h_2 \in F_N/F_{N-1}$, $c \in \mathbf{k}$, and let p_1, p_2 be the preimages under $\text{can}_{F_{N-1}}$ of h_1 and h_2 respectively. Since $\text{can}_{F_{N-1}}$ is $\mathbf{k}$ -linear it follows that

$$\text{can}_{F_{N-1}}(cp_1 + p_2) = c\text{can}_{F_{N-1}}(p_1) + \text{can}_{F_{N-1}}(p_2) = ch_1 + h_2,$$

hence $cp_1 + p_2 \in \text{can}_{F_{N-1}}^{-1}(ch_1 + h_2)$. Recalling that $\text{can}_{F_{N-1}} \circ \psi$ is $\mathbf{k}$ -linear it follows that

$$(\text{can}_{F_{N-1}} \circ \psi)(cp_1 + p_2) = c(\text{can}_{F_{N-1}} \circ \psi)(p_1) + (\text{can}_{F_{N-1}} \circ \psi)(p_2).$$

It now follows by definition of $(\text{gr}\psi)_N$ that

$$(\text{gr}\psi)_N(ch_1 + h_2) = c(\text{gr}\psi)_N(h_1) + (\text{gr}\psi)_N(h_2),$$

hence $(\text{gr}\psi)_N$ is a $\mathbf{k}$ -linear map. Note that since $N \in \mathbb{N}$ was arbitrary, it follows that the above holds for all $N \in \mathbb{N}$, and a nice way of summarising this is to say that the following diagram:

$$\begin{array}{ccc} F_N & \xrightarrow{\pi} & F'_N \\ \text{can}_{F_{N-1}} \downarrow & & \downarrow \text{can}_{F'_{N-1}} \\ F_N/F_{N-1} & \xrightarrow{(\text{gr}\pi)_N} & F'_N/F'_{N-1} \end{array}$$

of $\mathbf{k}$ -vector spaces and their morphisms commutes, for all $N \in \mathbb{N}$. We now collect these maps into a map $\text{gr}\psi : \text{gr}_F A \rightarrow \text{gr}_{F'} B$ which is defined for $h \in \text{gr}_F A$ by

$$\text{gr}\psi(h) = \text{gr}\psi\left(\sum_i h_i\right) = \sum_i (\text{gr}\psi)_i(h_i),$$

where we have used the grading in $\text{gr}_F A$ to write h as a sum of homogenous elements where $h_i \in F_i/F_{i-1}$ for all i . We note that since $(\text{gr}\psi)_N$ is $\mathbf{k}$ -linear for all $N \in \mathbb{N}$ it follows by definition of $\text{gr}\psi$ as a sum of $\mathbf{k}$ -linear maps that $\text{gr}\psi$ is $\mathbf{k}$ -linear map. Moreover, we claim that this is in fact an algebra morphism. By the work above, it suffices to prove that for any $h_1, h_2 \in \text{gr}_F A$ we have $\text{gr}\psi(h_1 h_2) = \text{gr}\psi(h_1) \text{gr}\psi(h_2)$, we first consider the case where both h_1, h_2 are homogenous and then generalise to arbitrary elements. Let $h_1 \in F_n/F_{n-1}$ and $h_2 \in F_m/F_{m-1}$, and let $p_1 \in F_n \setminus F_{n-1}$ and $p_2 \in F_m \setminus F_{m-1}$ be preimages for h_1 and h_2 respectively. Then we note that by definition of multiplication

in $\text{gr}_F A$ we have $h_1 h_2 \in F_{n+m}/F_{n+m-1}$ and furthermore

$$h_1 h_2 = p_1 p_2 + F_{n+m-1} = \text{can}_{F_{n+m-1}}(p_1 p_2),$$

hence $p_1 p_2 \in \text{can}_{F_{n+m-1}}^{-1}(h_1 h_2)$. From this we obtain

$$\begin{aligned} \text{gr}\psi(h_1 h_2) &= \text{can}_{F_{n+m-1}}(\psi(p_1 p_2)) \\ &= \text{can}_{F_{n+m-1}}(\psi(p_1)\psi(p_2)) \\ &= \text{can}_{F_{n-1}}(\psi(p_1))\text{can}_{F_{m-1}}(\psi(p_2)) \\ &= \text{gr}\psi(h_1)\text{gr}\psi(h_2), \end{aligned}$$

where we have used the assumption that ψ is an algebra morphism for the second equality, the definition of multiplication in $\text{gr}_{F'} B$ for the third equality noting that since ψ is filtered and by the assumptions $p_1 \in F_n$ and $p_2 \in F_m$ it follows that $\psi(p_1) \in F'_n$ and $\psi(p_2) \in F'_m$ hence both $\text{can}_{F_{n-1}}(\psi(p_1))$ and $\text{can}_{F_{m-1}}(\psi(p_2))$ make sense. Finally for the fourth equality we have used the definition of $\text{gr}\psi$ and the assumption that p_1, p_2 are the preimages of h_1 and h_2 respectively under the appropriate maps.

Having shown the desired result for homogenous elements of $\text{gr}_F A$, we now consider general elements $h, h' \in \text{gr}_F A$ which we can write as $h = \sum_i h_i$ and $h' = \sum_j h'_j$ where $h_i \in F_i/F_{i-1}$ for all i and $h'_j \in F_j/F_{j-1}$ for all j . We write the product

$$hh' = \left(\sum_i h_i \right) \left(\sum_j h'_j \right) = \sum_{i,j} h_i h'_j,$$

where we note that $h_i h'_j \in F_{i+j}/F_{i+j-1}$ is homogenous. Therefore we deduce

$$\text{gr}\psi(hh') = \sum_{i,j} \text{gr}\psi(h_i h'_j) = \sum_{i,j} \text{gr}\psi(h_i)\text{gr}\psi(h'_j) = \text{gr}\psi(h)\text{gr}\psi(h'),$$

where the first equality follows by linearity of $\text{gr}\psi$, and the second equality follows from our result about homogenous elements. It now follows that $\text{gr}\psi : \text{gr}_F A \rightarrow \text{gr}_{F'} B$ is an algebra morphism.

Given an algebra A , the associated graded algebra $\text{gr}_F A$ can vary wildly depending on the choice of filtration F . For example, let $A = \mathbf{k}[x]$ and let $F_\bullet$ be a filtration on A defined by $F_n = \mathbf{k} \cdot (x^i : 0 \leq i \leq n)$ for all $n \in \mathbb{N}$, and let $\Gamma_\bullet$ be a filtration on A defined by $\Gamma_0 = \Gamma_1 = \mathbf{k}$ and $\Gamma_n = \mathbf{k} \cdot (x^{i-1} : 1 \leq i \leq n)$ for all $n \in \mathbb{Z}_+$. We invite the reader to check that $F_\bullet$ and $\Gamma_\bullet$ are indeed filtrations on A . Then note that on one hand,

$$\text{gr}_F A = \bigoplus_{n \in \mathbb{N}} \frac{F_n}{F_{n-1}} = \frac{\mathbf{k}}{0} \oplus \frac{\mathbf{k} \cdot (1, x)}{\mathbf{k}} \oplus \frac{\mathbf{k} \cdot (1, x, x^2)}{\mathbf{k}(1, x)} \oplus \dots$$

and in fact one can check that $\text{gr}_F A \cong A$ in this case. On the other hand,

$$\text{gr}_\Gamma A = \bigoplus_{n \in \mathbb{N}} \frac{\Gamma_n}{\Gamma_{n-1}} = \frac{\mathbf{k}}{0} \oplus \frac{\mathbf{k}}{\mathbf{k}} \oplus \frac{\mathbf{k} \cdot (x)}{\mathbf{k}} \oplus \frac{\mathbf{k} \cdot (x, x^2)}{\mathbf{k} \cdot (x)} \oplus \dots$$

and by definition of $\Gamma_\bullet$ we have that $\overline{x^i} \in F_{i+1}/F_i$, from which it follows that for any $i, j \in \mathbb{Z}_+$

$$\overline{x^i} \cdot \overline{x^j} = x^i \cdot x^j + F_{i+j+1} = x^{i+j} + F_{i+j+1} = 0$$

since by definition of $\Gamma_\bullet$ $x^{i+j} \in F_{i+j+1}$. We now see that $\text{gr}_F A$ is Noetherian whilst $\text{gr}_\Gamma A$ is not even finitely generated! The above example suggests that trying to solve the MR question for S by considering *any* finite dimensional filtration of S will be a **hard** problem. To make progress, we will define a special class of finite dimensional filtrations which we call McConnell-Stafford filtrations (MS filtrations for short), and go on to show that these filtrations have some nice properties. Before we can do so, we need to go over some preliminaries.

Let $\mathbf{k}\langle X_1, \dots, X_n \rangle$ be the free algebra in n variables and let $d_1, \dots, d_n \in \mathbb{N}$. Since the free algebra has no relations it follows that if we define

$$T_m = \mathbf{k} \cdot (X_{i_1} \cdots X_{i_j} \mid \sum_j d_{i_j} = m, i_1, \dots, i_j \in \{1, \dots, n\}),$$

then $\mathbf{k}\langle X_1, \dots, X_n \rangle = \bigoplus_{j \in \mathbb{N}} T_j$ is a grading, where note that we define the 0-fold product as 1, i.e. $T_0 = \mathbf{k}$. Given a grading on a $\mathbf{k}$ -algebra $A = \bigoplus_j A_j$, we can construct a filtration $F_\bullet$ on A by defining $F_n = \bigoplus_{j=1}^n A_j$, hence by the above discussion we obtain a filtration $\mathcal{F}_\bullet$ on $\mathbf{k}\langle X_1, \dots, X_n \rangle$ by defining $\mathcal{F}_n = \bigoplus_{j=1}^n T_j$. If A is finitely generated by n generators, say $g_1, \dots, g_n \in A$, it follows by the universal property of free associative algebras that we have an algebra morphism $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ which is defined by $\pi(X_i) = g_i$ for all i . We wish to obtain a filtration on A given the filtration $\mathcal{F}_\bullet$ and the morphism π ; the following proposition will allow us to do precisely that.

Proposition 44. *Let $F_n := \pi(\mathcal{F}_n)$, then $F_\bullet$ is a filtration on A .*

Proof. We check that $F_\bullet$ satisfies the definition of a filtration.

1. fix some $n \in \mathbb{N}$, then $F_n = \pi(\mathcal{F}_n) \subseteq \psi(\mathcal{F}_{n+1}) = F_{n+1}$ where the inclusion follows since $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ since $\mathcal{F}_\bullet$ is a filtration and algebra morphisms preserve inclusions.
2. for some fixed $n, m \in \mathbb{N}$ we see that,

$$F_n F_m = \psi(\mathcal{F}_n) \psi(\mathcal{F}_m) = \psi(\mathcal{F}_n \mathcal{F}_m) \subseteq \psi(\mathcal{F}_{n+m}) = F_{n+m}$$

where we have used the assumption that π is an algebra morphism and that $\mathcal{F}_n \mathcal{F}_m \subseteq \mathcal{F}_{n+m}$ which follows since $\mathcal{F}_\bullet$ is a filtration.

3. We have $1 \in \mathcal{F}_0$; since π is an algebra morphism it follows that $1 \in \psi(\mathcal{F}_0) = F_0$.
4. Since $F_n = \pi(\mathcal{F}_n) \subseteq A$ for all $n \in \mathbb{N}$, it follows that $\bigcup_{n \in \mathbb{N}} F_n \subseteq A$. For the other inclusion let $f \in A$, then since π is surjective it follows that there exists $h \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that $\pi(h) = f$. Since $\mathcal{F}_\bullet$ is a filtration it follows that for some $m \in \mathbb{N}$ we have $h \in \mathcal{F}_m$, hence $f = \pi(h) \in \pi(\mathcal{F}_m) = F_m$ which immediately implies that $f \in \bigcup_{n \in \mathbb{N}} F_n$. Since this holds for arbitrary $f \in A$, the other inclusion follows and we have the result $\bigcup_{n \in \mathbb{N}} F_n = A$.

It now follows by definition that $F_\bullet$ is a filtration on A . $\square$

To define MS filtrations we will want to use a minimal number of elements $g_1, \dots, g_n \in A$ to generate A . We impose this constraint in order to exclude filtrations where products of generators do not have the F -degree we expect them to have. We illustrate this situation in the example below.

Example 45. For example, let $A = \mathbf{k}[x]$, let it be generated by $x, x^2 \in A$ and let $F_\bullet$ be defined as described above by putting a grading on the free algebra $\mathbf{k}\langle X_1, X_2 \rangle$ such that $\deg_F(x) = 2$ and $\deg_F(x^2) = 3 \neq 4$. Whilst A can be generated by x alone, and hence we have a surjective morphism (in fact an isomorphism) $\pi : \mathbf{k}\langle X_1 \rangle \rightarrow A$, one can see that we cannot put a filtration $\mathcal{F}_\bullet$ on $\mathbf{k}\langle X_1 \rangle$ using our grading procedure such that π is strict, since we would require that $X_1 \in \mathcal{F}_2 \setminus \mathcal{F}_1$ which would determine that $(X_1)^2 \in \mathcal{F}_4 \setminus \mathcal{F}_3$ and these two observations would imply that $\pi(X_1) = x \in F_2 \setminus F_1$ and $\pi((X_1)^2) = x^2 \in F_4 \setminus F_3$ respectively, which contradicts our assumption that $\deg_F(x^2) = 3$.

Recalling the language of Definition 36, we will refer to $\{g_1, \dots, g_n\} \subseteq A$ as a minimal generating set of A . Note that in the example above x minimally generates $\mathbf{k}[x]$, whilst x, x^2 does not minimally generate $\mathbf{k}[x]$.

4.3.2 Definition and Properties

We are now in the position to state the definition of MS filtrations.

Definition 46. Let A be a finitely generated, filtered $\mathbf{k}$ -algebra with filtration $F_\bullet$. Let $\{g_1, \dots, g_n\}$ be a minimal generating set of A , and let $d_i = \deg_F(g_i) \in \mathbb{Z}_+$. Define a grading on the free algebra $\mathbf{k}\langle X_1, \dots, X_n \rangle \oplus_{j \in \mathbb{N}} T_j$ by

$$T_m = \mathbf{k} \cdot (X_{i_1} \cdots X_{i_j} : \sum_j d_{i_j} = m, i_1, \dots, i_j \in \{1, \dots, n\}),$$

where note that we define the 0-fold product as 1, i.e. $T_0 = \mathbf{k}$. Define a filtration $\mathcal{F}_\bullet$ on the free algebra by $\mathcal{F}_n = \bigoplus_{j=1}^n T_j$ for all $j \in \mathbb{N}$. We say the filtration $F_\bullet$ on A is McConnell-Stafford if the surjective morphism $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ given by $\pi(X_i) = g_i$ is strict, i.e. $\pi(\mathcal{F}_n) = F_n$.

Remark 47. By the above definition we have that an MS filtration $F_\bullet$ on an algebra A comes equipped with a surjective and strict morphism, $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ such that

$\{\pi(X_1), \dots, \pi(X_n)\}$ is a minimal generating set of A . From now whenever we say that we have an MS filtration $F_\bullet$ on an algebra A with minimal generating set $\{g_1, \dots, g_n\}$, we will implicitly also say that we have the filtered free algebra $\mathbf{k}\langle X_1, \dots, X_n \rangle$ with filtration $\mathcal{F}_\bullet$ as defined in definition 46 and the strict and surjective morphism $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ given by $\pi(X_i) = g_i$. Moreover, the way we have defined the grading in Definition 46 is rather tedious to write out, hence we will say that we have a grading on the free algebra with $\deg(X_i) = d_i$ whenever we wish to define a grading as in the definition above.

We now make a small detour to discuss Bergman's Diamond Lemma which will give us a preferred basis $\mathcal{B}$ of S which will behave well under the gr_F map. Some good resources on the Diamond Lemma include [Ber78] and Section 1.4 of [Rog14]. We will take for granted that the reader is familiar with reduction systems, and the process of resolving ambiguities, and proceed to state (without proof) the results and apply them. The Diamond Lemma may be partially stated as follows.

Theorem 48 (Diamond Lemma). *Let $\mathbf{k}\langle X_1, \dots, X_n \rangle$ be a free associative algebra and let I be an ideal generated by $\{r_\sigma\}_{\sigma \in \mathcal{I}}$, for some indexing set $\mathcal{I}$ where $\forall \sigma \in \mathcal{I}$, $R_\sigma = \omega_\sigma - f_\sigma$, where ω_σ is the leading word of r_σ under some admissible ordering $\preceq$ of the monoid $\langle X_1, \dots, X_n \rangle$. The following are equivalent:*

1. *all ambiguities among r_σ are resolvable with respect to $\preceq$,*
2. *Let $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow \mathbf{k}\langle X_1, \dots, X_n \rangle/I$ be the canonical projection, and let $\langle X \rangle_{\text{irr}}$ be the set of irreducible words. Then $\pi(\langle X \rangle_{\text{irr}})$ is a $\mathbf{k}$ -basis of $\mathbf{k}\langle X_1, \dots, X_n \rangle/I$.*

Proof. See proof of Theorem 1.2 in [Ber78]. □

Remark 49. We will use terms “word” and “monomial” interchangeably, noting that any non-zero monomial can be appropriately rescaled (since all algebras are over a field $\mathbf{k}$) into monic monomials of $\mathbf{k}\langle X_1, \dots, X_n \rangle$ which are precisely words of the monoid $\langle X_1, \dots, X_n \rangle$.

Let A be a filtered $\mathbf{k}$ -algebra with a MS filtration $F_\bullet$ and let A be minimally generated by $g_1, \dots, g_n$. Let $d_i = \deg_F(g_i)$ and consider the free algebra $\mathbf{k}\langle X_1, \dots, X_n \rangle$ along side the degree-lexicographic ordering $\preceq$ on the monoid $\langle X_1, \dots, X_n \rangle$, defined by $\deg(X_i) = d_i$ and $X_1 \preceq \dots \preceq X_n$. Note that we have the canonical surjection $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ given by $\pi(X_i) = g_i$ such that defining $I := \ker(\pi)$ we can write $A \cong \mathbf{k}\langle X_1, \dots, X_n \rangle/I$ using the first isomorphism theorem. Moreover, let $\{r_\sigma\}_{\sigma \in \mathcal{I}}$ be a generating set for the ideal I , where $\mathcal{I}$ is some indexing set. Then by the Diamond Lemma, if all ambiguities among r_σ are resolvable with respect to $\preceq$, then $\pi(\langle X \rangle_{\text{irr}})$ is a $\mathbf{k}$ -basis for A . We denote this basis $\mathcal{B} := \pi(\langle X \rangle_{\text{irr}})$. We will often use ω to denote a word in the free algebra, and we will denote its image under π as $\omega(\underline{g}) := \pi(\omega)$ where $\underline{g}$ in the notation refers to the choice of minimal generating set $\{g_1, \dots, g_n\}$ of A . $\pi : \langle X \rangle_{\text{irr}} \rightarrow \pi(\langle X \rangle_{\text{irr}}) = \mathcal{B}$ is clearly surjective, but moreover if for $\omega, \omega' \in \langle X \rangle_{\text{irr}}$ we would have that $\pi(\omega) = \pi(\omega')$ then we could construct the non-trivial relation $\pi(\omega) - \pi(\omega') = 0$ in A with which we

could either reduce ω or ω' , which would contradict the assumption that both ω and ω' are irreducible words. It follows that we have a bijection between $\langle X \rangle$ and $\mathcal{B}$, hence we can write any element of $\mathcal{B}$ as $\omega(\underline{g})$ where ω is the unique element of $\langle X \rangle_{\text{irr}}$ such that $\pi(\omega) = \omega(\underline{g})$. We now connect the ordering $\preceq$ with the filtration $\mathcal{F}_\bullet$ in the following lemma,

Lemma 50. *Let $\omega, \omega' \in \langle X_1, \dots, X_n \rangle$ be monomials. If $\deg_{\mathcal{F}}(\omega) < \deg_{\mathcal{F}}(\omega')$ then $\omega \preceq \omega'$.*

Proof. By assumption $\deg_{\mathcal{F}}(\omega) < \deg_{\mathcal{F}}(\omega')$ and note that the ordering $\preceq$ uses the same grading on the free algebra $\mathbf{k}\langle X_1, \dots, X_n \rangle$ to order monomials, that is used to define the filtration $\mathcal{F}_\bullet$. What this means precisely is that $\deg_{\mathcal{F}}$ and $\deg$ coincide, where we think of $\deg$ as the degree function associated to the grading on the free algebra. Hence our assumption implies that $\deg(\omega) < \deg(\omega')$ and so by the definition of the degree-lexicographic ordering $\preceq$, it now follows that $\omega \preceq \omega'$. $\square$

With the above lemma we are now ready to proceed to show how elements of $\mathcal{B}$ interact well with the gr map, captured in the following results.

Lemma 51. *Let A be an algebra with an MS-filtration $F_\bullet$ given by the minimal generators $g_1, \dots, g_n$. Let $\omega \in \langle X \rangle_{\text{irr}}$, then $\deg_F(\omega(\underline{g})) = \deg_{\mathcal{F}}(\omega)$.*

Proof. Let $\deg_{\mathcal{F}}(\omega) = N$ and $\deg_{\mathcal{F}}(\omega(\underline{g})) = M$ for some $N, M \in \mathbb{N}$. By definition we have a strict algebra morphism $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$, meaning that for all n , $\pi(\mathcal{F}_n) = F_n$, hence $\pi(\mathcal{F}_N) \subseteq F_N$ which implies that $\pi(\omega) = \omega(\underline{g}) \in F_N$ and so we have $M \leq N$. Now assume for a contradiction that $M < N$, since $F_M \subseteq \pi(\mathcal{F}_M)$, we have that $\exists h \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that $h \in \pi^{-1}(\omega(\underline{g})) \cap \mathcal{F}_N$. Write $h = \omega' - f$ where ω' is the leading word under $\preceq$, note that since $\preceq$ and $\mathcal{F}_\bullet$ use the same grading on the free algebra in their definitions it follows that $\deg_{\mathcal{F}}(\omega') = \deg_{\mathcal{F}}(h) = M$. Therefore $\deg_{\mathcal{F}}(\omega') < \deg_{\mathcal{F}}(\omega)$ which by Lemma 50 implies $\omega' \preceq \omega$, where note that by assumption $\omega' \neq \omega$ since they have different $\mathcal{F}$ -degrees. This implies that ω can be further reduced with respect to the relations generating the kernel of π , however that contradicts our assumption that $\omega \in \langle X \rangle_{\text{irr}}$ which implies that ω is an irreducible word. It follows that $M \geq N$. We have now shown both inequalities, and so it follows that $\deg_{\mathcal{F}}(\omega) = \deg_F(\omega(\underline{g}))$. $\square$

Lemma 52. *With notation as above, let $\omega \in \mathbf{k}\langle X \rangle_{\text{irr}}$. Then $\text{gr}_F(\omega(\underline{g})) = \omega(\text{gr}_F(\underline{g}))$.*

Proof. By Lemma 51 $\deg_{\mathcal{F}}(\omega) = \deg_F(\omega(\underline{g})) = N$ for some $N \in \mathbb{N}$. Let $\omega(\underline{X}) = \prod_{l=1}^r X_{i_l}$, therefore $\omega(\underline{g}) = \prod_{l=1}^r g_{i_l}$ and by the definition of $\mathcal{F}_\bullet$ we have that $N = \sum_{l=1}^r d_{i_l}$ where $d_{i_l} = \deg_{\mathcal{F}}(X_{i_l}) = \deg_F(g_{i_l})$. We note that

$$\omega(\text{gr}_F(\underline{g})) = \prod_{l=1}^r \text{gr}_F(g_{i_l}) \in F_{\sum_{l=1}^r d_{i_l}} / F_{\sum_{l=1}^r d_{i_l} - 1} = F_N / F_{N-1},$$

and since we have $\deg_F(\omega(\underline{g})) = N$ it follows that $\text{gr}_F(\omega(\underline{g})) \in F_N / F_{N-1}$. We have shown that $\omega(\text{gr}_F(\underline{g}))$ and $\text{gr}_F(\omega(\underline{g}))$ belong to the same graded component of $\text{gr}_F A$,

hence by the definition of multiplication in $\text{gr}_F A$ we have equality between them. Hence $\omega(\text{gr}_F(\underline{g})) = \text{gr}_F(\omega(\underline{g}))$. $\square$

Lemma 53. *With notation as above, let $p \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that when writing $p = \sum_i c_i \omega_i$ where $c_i \in \mathbf{k}$ and $\omega_i \in \langle X \rangle_{\text{irr}}$ it follows that for all i , $\deg_{\mathcal{F}}(\omega_i) = N$ for some $N \in \mathbb{N}$. Then $\text{gr}_F(p(\underline{g})) = p(\text{gr}_F(\underline{g}))$.*

Proof. First note that $\forall n \in \mathbb{N}$, gr_F is a $\mathbf{k}$ -linear map on $F_n \setminus F_{n-1}$ and by assumption for all i , we have $\omega_i \in \langle X \rangle_{\text{irr}}$ and $\deg_{\mathcal{F}}(\omega_i) = N$, so by Lemma 51 it follows that $\omega_i(\underline{g}) \in F_N \setminus F_{N-1}$ for all i . Furthermore, by our choice of grading with respect to $X_1, \dots, X_n$ in the definition of $\mathcal{F}_{\bullet}$ it follows that $\deg_{\mathcal{F}}(p) = \max_i \{\deg_{\mathcal{F}}(\omega_i)\} = N$, since for all i we have $\deg_{\mathcal{F}}(\omega_i) = N$. It now follows that,

$$\begin{aligned}
 & \text{gr}_F(p(\underline{g})) \\
 &= \text{gr}_F \left(\sum_i c_i \omega_i(\underline{g}) \right) && \text{by definition of } p, \\
 &= \sum_i c_j \text{gr}_F(\omega_i(\underline{g})) && \text{since gr is } \mathbf{k}\text{-linear on } F_N \setminus F_{N-1}, \\
 &= \sum_i c_j \omega_i(\text{gr}_F(\underline{g})) && \text{by Lemma 52 and } \omega_i \in \langle X \rangle_{\text{irr}}, \forall j, \\
 &= p(\text{gr}_F(\underline{g})) && \text{by definition of } p.
 \end{aligned}$$

$\square$

Lemma 54. *With notation as above, let $p = \sum_i c_i \omega_i \in A$ where $c_i \in \mathbf{k}$ and $\omega_i \in \langle X \rangle_{\text{irr}}$ for all i . If $N = \max_i \{\deg_F(\omega_i(\underline{g}))\}$, then $\deg_F(p) = N$.*

Proof. By assumption $\omega_i \in \langle X \rangle_{\text{irr}}$, hence by Lemma 51 it follows that for all i $\deg_F(\omega_i(\underline{g})) = \deg_{\mathcal{F}}(\omega_i)$. By our choice of grading with respect to $X_1, \dots, X_n$ in the definition of $\mathcal{F}_{\bullet}$, since $p = \sum_i c_i \omega_i$, we have that

$$\deg_{\mathcal{F}}(p) = \max_i \{\deg_{\mathcal{F}}(\omega_i)\} = \max_i \{\deg_F(\omega_i(\underline{g}))\} = N,$$

where the second equality comes from our observation that $\deg_F(\omega_i(\underline{g})) = \deg_{\mathcal{F}}(\omega_i)$ for all i , and the third equality follows by assumption. By strictness of π we have that $\pi(\mathcal{F}_N) = F_N$ which together with $\deg_{\mathcal{F}}(p) = N$ implies that $\deg_F(p(\underline{g})) \leq N$. Now assume for a contradiction that $\deg_F(p(\underline{g})) = M < N$, then by the strictness of π , $\exists h \in \mathcal{F}_M \cap \pi^{-1}(p(\underline{g}))$ and write $h = \sum_j c'_j \omega_j$ where $c'_j \in \mathbf{k}$ and ω_j is a word in the free algebra, for all j . Let ω denote the leading word of p , and note that $\deg_{\mathcal{F}}(\omega) = N$. Since $\deg_{\mathcal{F}}(h) = M$ it follows by the grading in the definition of $\mathcal{F}_{\bullet}$ that $\deg_{\mathcal{F}}(\omega'_j) \leq M < N$ for all j , hence by Lemma 50 it follows that for all j , we have $\omega'_j \preceq \omega$. Observe that $p - h \in \ker \pi$, hence this is a non-trivial relation with leading word ω , which implies that $\omega(\underline{X})$ is not irreducible, however by assumption $\omega \in \langle X \rangle_{\text{irr}}$, i.e. ω is irreducible: contra-

diction. It thus follows that $\deg_F(p(\underline{g})) \geq N$, and having now showed both inequalities it follows that $\deg_F(p(\underline{g})) = N$. $\square$

Lemma 55. *With notation as above, for some $N \in \mathbb{N}$, let $h \in F_N/F_{N-1}$. Then $\exists p \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that $\deg_F(p(\underline{g})) = N$, $\text{gr}_F(p(\underline{g})) = h$, and $p = \sum_i c_i \omega_i$ where $\omega_i \in \langle X \rangle_{\text{irr}}$, $c_i \in \mathbf{k}$ and $\deg_F(\omega_i(\underline{g})) = N$ for all i .*

Proof. First note that by definition gr_F is surjective, hence $\exists q \in A$ such that $\deg_F(q) = N$ and $\bar{q} = h$. We write q in terms of the basis $\mathcal{B}$ as $q = \sum_i c_i \omega_i(\underline{g})$ where $c_i \in \mathbf{k}$ and $\omega_i(\underline{g}) \in \mathcal{B}$ for all i . Note that by Lemma 54 it follows that $\deg_F(\omega_i(\underline{g})) \leq N$ for all i , as otherwise this lemma would imply that $\deg_F(q) > N$ which contradicts our choice of q . Let $L = \{i : \deg_F(\omega_i(\underline{g})) < N\}$, and let $q' = \sum_{j \in L} c_j \omega_j(\underline{g})$, by Lemma 54 since $\deg_F(\omega_j(\underline{g})) < N$ for all $j \in L$ it follows that $\deg_F(q') < N$ and moreover $\deg_F(q - q') = N$. In fact, there exists an index α such that $\deg_F(\omega_\alpha) = N$, as otherwise $i \in L$ for all i , which would imply by the above that $\deg_F(q) < N$, which contradicts our choice of q . Let $\text{can}_{F_{N-1}}$ denote the canonical surjection $F_N \rightarrow F_N/F_{N-1}$, then since $\deg_F(q - q') = N$ we have

$$\text{gr}_F(q - q') = \text{can}_{F_{N-1}}(q - q') = \text{can}_{F_{N-1}}(q) - \text{can}_{F_{N-1}}(q') = h - 0 = h,$$

where we note that the second equality follows since $\text{can}_{F_{N-1}}$ is an algebra morphism, and the third equality follows since $\deg_F(q') < N$ and since by assumption $\bar{q} = h$. It follows that by defining $p = \sum_{i \notin L} c_i \omega_i$ so that $p(\underline{g}) = q - q'$, we have that $\deg_F(p(\underline{g})) = N$, $\text{gr}_F(p(\underline{g})) = h$ and $\deg_F(\omega_i(\underline{g})) = N$ for all i , as required. $\square$

We will also require the following result below,

Proposition 56. *Let $\mathbf{k}\langle X_1, \dots, X_n \rangle$ be the filtered free algebra with filtration $\mathcal{F}_\bullet$ defined by $\mathcal{F}_n = \mathbf{k} \cdot (X_{i_1} \cdots X_{i_t} \mid \sum_{j=1}^t d_{i_j} \leq n)$, where $d_{i_j} := \deg_{\mathcal{F}}(X_{i_j})$. Then $\mathbf{k}\langle X_1, \dots, X_n \rangle \cong \text{gr}_{\mathcal{F}} \mathbf{k}\langle X_1, \dots, X_n \rangle$, where the isomorphism of $\mathbf{k}$ -algebras is given by $X_i \mapsto \bar{X}_i$.*

Proof. We begin by pointing out that $\mathcal{F}_\bullet$ is an MS filtration on the free algebra where $\pi = \text{id}$. By the universal property of free associative algebras we have a morphism $\theta : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow \text{gr}_{\mathcal{F}} \mathbf{k}\langle X_1, \dots, X_n \rangle$ given by $\theta(X_i) = \bar{X}_i$, we claim that θ is an isomorphism. First surjectivity, let $f \in \text{gr}_{\mathcal{F}} \mathbf{k}\langle X_1, \dots, X_n \rangle$ and write $f = \sum_j f_j$ where $f_j \in \mathcal{F}_j/\mathcal{F}_{j-1}$. For each f_j , by Lemma 55 we have $p_j \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that $\deg_{\mathcal{F}}(p_j) = j$, $\bar{p}_j = f_j$ and $p_j = \sum_i c_i \omega_i$ where $\omega_i \in \langle X \rangle_{\text{irr}}$ and $\deg_{\mathcal{F}}(\omega_i) = j$ for all i . Now we may apply Lemma 53 to conclude that $f = \text{gr}_{\mathcal{F}}(p_j(\underline{X})) = p_j(\text{gr}_{\mathcal{F}}(\underline{X}))$. Therefore we can write each homogenous component of f as a polynomial in $\bar{X}_1, \dots, \bar{X}_n$, and so doing this for each f_j and taking the sum we see that f is a polynomial in $\bar{X}_1, \dots, \bar{X}_n$, i.e. $f = \theta(\sum_j p_j)$ which shows that θ is surjective.

Now for injectivity, let $q \in \mathbf{k}\langle \underline{X} \rangle$ such that $\theta(q) = 0$, and write $q = \sum_j c_j \omega_j(\underline{X})$ where $\omega_j \in \langle X \rangle_{\text{irr}}$. By assumption we have

$$0 = \sum_j c_j \omega_j(\text{gr}_{\mathcal{F}}(\underline{X})) = \sum_i \sum_l c_{il} \omega_{il}(\text{gr}_{\mathcal{F}}(\underline{X})),$$

where $\sum_l \gamma_{il} \omega_{il}(\text{gr}_{\mathcal{F}}(\underline{X}))$ is a homogenous component of degree i . Using the grading of the associated graded algebra, we deduce that $\sum_l c_{il} \omega_{il}(\text{gr}_{\mathcal{F}}(\underline{X})) = 0$ for all i .

Now, fix some i and assume that some $c_{il} \neq 0$, we have by assumption that $\deg_{\mathcal{F}}(\sum_l c_{il} \omega_{il}(\underline{X})) < i$. By Lemma 52, $\text{gr}_{\mathcal{F}}(\omega_{il}(\underline{X})) = \omega_{il}(\text{gr}_{\mathcal{F}}(\underline{X}))$ and since each $\omega_{il}(\text{gr}_{\mathcal{F}}(\underline{X})) \in \mathcal{F}_i / \mathcal{F}_{i-1}$ it follows that $\deg_{\mathcal{F}}(\omega_{il}(\underline{X})) = i$. Hence by the grading used in the definition of $\mathcal{F}_{\bullet}$ it follows that $\deg_{\mathcal{F}}(\sum_l \gamma_{il} \omega_{il}(\underline{X})) = i$, which contradicts $\sum_l \gamma_{il} \omega_{il}(\underline{X}) \in \mathcal{F}_{i-1}$. It now follows that for our choice of i , all $c_{il} = 0$, and since i was arbitrary it follows that $c_{il} = 0$ for all i, l , which implies that $q = 0$, hence $\ker(\theta) = 0$ and we have injectivity. $\square$

The results above show that given a filtered algebra A with an MS filtration $F_{\bullet}$, minimally generated by $g_1, \dots, g_n \in A$, we can use the Diamond Lemma to construct a basis $\mathcal{B}$ of A which behaves nicely with the map gr_F . Using this observation we prove the first nice property of MS filtrations in the proposition below.

Proposition 57. *let A be a filtered $\mathbf{k}$ -algebra with a MS filtration $F_{\bullet}$, and let A be minimally generated by $g_1, \dots, g_n \in A$. Then $\text{gr}_F A$ is finitely generated by $\bar{g}_1, \dots, \bar{g}_n$.*

Proof. Recall from Construction 43 that π induces the algebra morphism $\text{gr}\pi : \text{gr}_{\mathcal{F}}\mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow \text{gr}_F A$, which we now show is surjective. By the definition of $\text{gr}\pi$ and the graded structure of both associated graded algebras, we remark that to show $\text{gr}\pi$ is surjective it is sufficient to prove $(\text{gr}\pi)_m$ is surjective for all $m \in \mathbb{N}$. Let us fix $m \in \mathbb{N}$, and recall from Construction 43 that $\text{gr}\pi$ satisfies the following commutative diagram,

$$\begin{array}{ccc} \mathcal{F}_m & \xrightarrow{\pi} & F_m \\ \text{can}_{\mathcal{F}_{m-1}} \downarrow & & \downarrow \text{can}_{F_{m-1}} \\ \mathcal{F}_m / \mathcal{F}_{m-1} & \xrightarrow{(\text{gr}\pi)_m} & F_m / F_{m-1} \end{array}$$

Since π is a strict morphism, it follows by definition that $\pi(\mathcal{F}_m) = F_m$ and since π as well as $\text{can}_{F_{m-1}}$ are surjective, it follows that the composition $\text{can}_{F_{m-1}} \circ \pi$ is also surjective. Let $p \in F_m / F_{m-1}$, then since $\text{can}_{F_{m-1}} \circ \pi$ is surjective it follows that $\exists q \in \mathcal{F}_m$ such that $(\text{can}_{F_{m-1}} \circ \pi)(q) = p$. Since the diagram above commutes it follows that,

$$p = (\text{can}_{F_{m-1}} \circ \pi)(q) = ((\text{gr}\pi)_m \circ \text{can}_{\mathcal{F}_{m-1}})(q) = (\text{gr}\pi)_m(\text{can}_{\mathcal{F}_{m-1}}(q)) = (\text{gr}\pi)_m(p'),$$

where $p' = \text{can}_{\mathcal{F}_{m-1}}(q) \in \mathcal{F}_m / \mathcal{F}_{m-1}$. Hence $\forall p \in F_m / F_{m-1}$, $\exists p' \in \mathcal{F}_m / \mathcal{F}_{m-1}$ such that $(\text{gr}\pi)_m(p') = p$, thus $(\text{gr}\pi)_m$ is surjective and so by our previous remarks in fact $\text{gr}\pi$ is surjective. By Proposition 56 we have $\text{gr}_F \mathbf{k}\langle X_1, \dots, X_n \rangle \cong \mathbf{k}\langle X_1, \dots, X_n \rangle$,

it follows that there is a surjective $\mathbf{k}$ -algebra homomorphism from $\mathbf{k}\langle X_1, \dots, X_n \rangle$ to $\text{gr}_F S$, which is equivalent to the condition that $\text{gr}_F S$ is finitely generated by the images $(\text{gr}\pi \circ \text{gr}_{\mathcal{F}})(X_1), \dots, (\text{gr}\pi \circ \text{gr}_{\mathcal{F}})(X_n)$. By the definition of $\text{gr}\pi$ we note that $(\text{gr}\pi \circ \text{gr}_{\mathcal{F}})(X_i) = (\text{gr}_F \circ \pi)(X_i) = \bar{g}_i$, which combined with the above shows that $\text{gr}_F A$ is finitely generated by $\bar{g}_1, \dots, \bar{g}_n$. $\square$

Now that we have a basis $\mathcal{B}$ such that $\forall \omega(g) \in \mathcal{B}$ satisfies $\text{gr}_F(\omega(g)) = \omega(\text{gr}_F(g))$ one may hope that the set $\text{gr}_F \mathcal{B} := \{\text{gr}_F(\omega(g)) \mid \omega \in \langle X \rangle_{\text{irr}}\}$, forms a basis of $\text{gr}_F A$. The remarkable thing is that this is precisely what happens, which we show in the theorem below.

Theorem 58. *Let A be a filtered algebra with MS filtration $F_\bullet$, minimally generated by $g_1, \dots, g_n$. Let $\mathcal{B}$ be the basis of A obtained through the Diamond Lemma using a degree-lexicographic ordering compatible with the filtration $F_\bullet$. Then $\text{gr}_F \mathcal{B}$ as defined above is a basis for $\text{gr}_F A$.*

Proof. We need to show that $\text{gr}_F \mathcal{B}$ spans $\text{gr}_F A$ and that the elements of $\text{gr}_F \mathcal{B}$ are linearly independent over $\mathbf{k}$. Define $\mathcal{B}_n = \mathcal{B} \cap (F_n \setminus F_{n-1})$ and let $\text{gr}_F \mathcal{B}_n$ denote the image of $\mathcal{B}_n$ under the map gr_F . We note that since $\forall \omega(g) \in \mathcal{B}_n$ by definition we have $\deg_F(\omega(g)) = n$, then $\text{gr}_F \mathcal{B}_n \subseteq F_n/F_{n-1}$, hence by the direct sum structure of $\text{gr}_F A$ it follows that for $i, j \in \mathbb{N}$ such that $i \neq j$ we have $\text{gr}_F \mathcal{B}_i \cap \text{gr}_F \mathcal{B}_j = \emptyset$. This implies that $\text{gr}_F \mathcal{B} = \bigsqcup_{n \in \mathbb{N}} \text{gr}_F \mathcal{B}_n$. By the above it is sufficient to show that for arbitrary $N \in \mathbb{N}$, $\text{gr}_F \mathcal{B}_N$ is a basis of F_N/F_{N-1} .

We begin by showing that $\text{gr}_F \mathcal{B}_N$ spans F_N/F_{N-1} . We have already seen that $\text{gr}_F \mathcal{B}_N \subseteq F_N/F_{N-1}$ and since F_N/F_{N-1} is a $\mathbf{k}$ subspace it follows that the span $\mathbf{k} \cdot (\text{gr}_F \mathcal{B}_N) \subseteq F_N/F_{N-1}$. For the other inclusion, let $h \in F_N/F_{N-1}$, by Lemma 55 $\exists p \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ such that $\deg_F(p(g)) = N$, $\text{gr}_F p(g) = h$ and $p = \sum_i c_i \omega_i(g)$ with $\deg_F(\omega_i(g)) = N$ where $\omega_i \in X_{\text{irr}}$. We have now satisfied the hypothesis of Lemma 53 hence it follows that

$$h = \text{gr}_F(p(g)) = p(\text{gr}_F(g)) = \sum_i c_i \omega_i(\text{gr}_F(g)),$$

where (using Lemma 52) $\omega_i(\text{gr}_F(g)) = \text{gr}_F(\omega_i(g)) \in \text{gr}_F \mathcal{B}_N$. It now follows that $h \in \mathbf{k} \cdot (\text{gr}_F \mathcal{B}_N)$ and since h was arbitrary it follows that $\mathbf{k} \cdot (\text{gr}_F \mathcal{B}_N) \subseteq F_N/F_{N-1}$. Having shown both inclusions, we obtain $\mathbf{k} \cdot (\text{gr}_F \mathcal{B}_N) = F_N/F_{N-1}$.

For linear independence, let $0 = \sum_i c_i \text{gr}_F(\omega_i(g))$ where $\text{gr}_F(\omega_i(g)) \in \text{gr}_F \mathcal{B}_N$ for all i . Since each summand is in the F_N/F_{N-1} graded component, on which the map gr_F is $\mathbf{k}$ -linear, it follows that $0 = \text{gr}_F(\sum_i c_i \omega_i(g))$, which implies that $\sum_i c_i \omega_i(g) \in F_{N-1}$. Now assume for a contradiction that some $c_i \neq 0$, by Lemma 54 we have that $\deg_F(\sum_i c_i \omega_i(g)) = \max_i \{\deg_F(\omega_i(g))\}$, since by assumption for all i we have $\text{gr}_F(\omega_i(g)) \in \text{gr}_F \mathcal{B}_N$ it follows that for all i we have $\omega_i(g) \in \mathcal{B}_N$. Therefore $\deg_F(\omega_i(g)) \in \mathcal{B}_N$ for all i , which together with the above gives $\deg_F(\sum_i c_i \omega_i(g)) = N$, which clearly contradicts our earlier observation that $\sum_i c_i \omega_i(g) \in F_{N-1}$. It now follows that for all i ,

$c_i = 0$, hence the linear combination we started with is trivial and thus follows linear independence of elements in $\text{gr}_F \mathcal{B}_N$. $\square$

For the next result we note that given a grading on the free algebra $\mathbf{k}\langle X_1, \dots, X_n \rangle = \bigoplus_j T_j$, in a manner akin to Proposition 44, we can use a surjection $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow A$ to obtain a grading on A , as $A = \bigoplus_j \pi(T_j)$. We capture this in the following lemma.

Lemma 59. *Let $\mathbf{k}\langle X_1, \dots, X_n \rangle = \bigoplus_j T_j$ be a grading on the free algebra, and let I be an ideal of $\mathbf{k}\langle X_1, \dots, X_n \rangle$ generated by homogenous elements $r_1, \dots, r_l$. Then $\mathbf{k}\langle X_1, \dots, X_n \rangle / I = \bigoplus_j (T_j / I_j)$ is a grading, where $I_j = I \cap T_j$.*

Proof. Since I is generated by homogenous elements, it follows that $I = \bigoplus_j I_j$ is a grading where $I_j = I \cap T_j$. To see this, let $p \in I$, we see that,

$$p = \sum_i \lambda_i r_i \rho_i = \sum_{i,n} \lambda_{i,n} r_i \rho_{i,n}$$

where $\lambda_i, \rho_i \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ and $\lambda_{i,n}, \rho_{i,n} \in \mathbf{k}\langle X_1, \dots, X_n \rangle$ are homogenous for all i, n . It follows that $\lambda_{i,n} r_i \rho_{i,n} \in I$, since $r_i \in I$, and so we have expressed $p \in I$ as a sum of elements of I which are homogenous elements with respect to the grading T_j , hence I is graded as above. Let $\text{can} : \mathbf{k}\langle X_1, \dots, X_n \rangle \rightarrow \mathbf{k}\langle X_1, \dots, X_n \rangle / I$ be the canonical surjection, then we claim that $\mathbf{k}\langle X_1, \dots, X_n \rangle / I = \bigoplus_j \text{can}(T_j)$. Since can is surjective it is clear that $\mathbf{k}\langle X_1, \dots, X_n \rangle / I = \sum_j \text{can}(T_j)$, and to justify the direct sum structure we fix some $i, j \in \mathbb{N}$ such that $i \neq j$ and let $h \in \text{can}(T_i) \cap \text{can}(T_j)$. It follows that $\exists p \in T_i$ and $q \in T_j$ such that $\text{can}(p) = h = \text{can}(q)$, hence for some $f \in I$ we may write $p = f + q$. Since I is graded, it follows that we can write $f = \sum_k f_k$ where each $f_k \in I_k$, and so we have $p = \sum_k f_k + q$. Comparing the homogenous components of this equality we deduce that $p = f_i \in I_i$ and $q = f_j \in I_j$, which implies that $\text{can}(p) = \text{can}(q) = 0$, hence $h = 0$ which implies that $\text{can}(T_i) \cap \text{can}(T_j) = 0$. Technically, we should show that an arbitrary finite intersection of $\text{can}(T_i)$ is empty, however one can sensibly generalise the above argument to the general case and we will leave this as an exercise for the curious reader. We have now shown that $\mathbf{k}\langle X_1, \dots, X_n \rangle / I = \bigoplus_j \text{can}(T_j)$ and the result now follows by noting that $\text{can}(T_j) \text{can}(T_{j'}) = \text{can}(T_j T_{j'}) \subseteq \text{can}(T_{j+j'})$, since can is an algebra morphism and $\mathbf{k}\langle X_1, \dots, X_n \rangle \cong \bigoplus_j T_j$ is a grading, and observing that by definition $\text{can}(T_j) = T_j / (I \cap T_j) = T_j / I_j$. $\square$

We quickly recall the first, second and fourth isomorphism theorems for modules, without proof, which we will use in later proofs in this section. A good reference for these results is Chapter 10 of [DF04], from which we take the statement of the isomorphism theorems. Note that the lemma below will be stated for left modules, however the analogous statements hold for right modules as well.

Lemma 60 (Isomorphism Theorems for Modules). *Let R be an arbitrary ring,*

- (First Isomorphism Theorem) *Let M, N be left R -modules and let $\varphi : M \rightarrow N$ be a R -module homomorphism. Then $\ker(\varphi)$ is a submodule of M and $M / \ker(\varphi) \cong \text{im}(\varphi)$.*

- (Second Isomorphism Theorem) Let A, B be submodules of the left R -module M . Then $(A + B)/B \cong A/(A \cap B)$.
- (Fourth Isomorphism Theorem) Let N be a submodule of the left R -module M . There is a bijection between the submodules of M which contain N and submodules of M/N . This correspondence is given by $A \leftrightarrow A/N$ for all submodules A of M which contain N . Moreover, this correspondence commutes with sums and intersections (i.e. is a lattice isomorphism between the lattice of submodules of M/N and the lattice of submodules of M which contain N).

Proof. See Theorem 4 in Section 10.2 of [DF04] □

Remark 61. The Fourth Isomorphism Theorem is also known as in the literature as the Correspondence Theorem.

Now we prove a quick technical lemma.

Lemma 62. Let R be a ring, let M, N be left (or right) R -modules and let $A \subseteq M$, $B \subseteq N$ be submodules. Then $(M \oplus N)/(A \oplus B) \cong (M/A) \oplus (N/B)$.

Proof. Let $c_A : M \rightarrow M/A$ and $c_B : N \rightarrow N/B$ be the canonical projections, and define the map $\lambda : M \oplus N \rightarrow (M/A) \oplus (N/B)$ by $\lambda((m, n)) = (c_A(m), c_B(n))$ for $m \in M$ and $n \in N$. The maps c_A, c_B are well-defined from which it follows that λ is well-defined. Let $(f, h) \in \ker(\lambda)$ for some $(f, h) \in M \oplus N$, we see that,

$$\begin{aligned}
 \lambda((f, h)) &= 0 \\
 \iff c_A(f) &= 0 \text{ and } c_B(h) = 0 \\
 \iff f &\in A \text{ and } h \in B \\
 \iff (f, h) &\in A \oplus B
 \end{aligned}$$

it follows that $\ker(\lambda) = A \oplus B$ and thus by the First Isomorphism Theorem, remarking that since c_A, c_B are surjective then so is λ , we see that,

$$\frac{M \oplus N}{A \oplus B} = \frac{M \oplus N}{\ker(\lambda)} \cong \text{im}(\lambda) = \frac{M}{A} \oplus \frac{N}{B}$$

□

Onto the proposition of interest.

Proposition 63. Let A be a filtered algebra with MS filtration $F_\bullet$, and let A be minimally generated by $g_1, \dots, g_n$. Let $\mathbf{k}\langle X_1, \dots, X_n \rangle$ be bigraded by $\deg(X_i) = (d_i, e_i) \in \mathbb{N}^2$, so that $\ker(\pi)$ is generated by e -homogenous elements, and so that the $\deg_{\mathcal{F}}(X_i) = d_i$. Then $gr_F A$ is bigraded.

Proof. For the sake of conciseness, unless otherwise stated, the sums (both $\oplus$ and $\sum$) will be taken over $\mathbb{N}$. First we point out that the bigrading on the free algebra is

equivalent to saying that $\mathbf{k}\langle X_1, \dots, X_n \rangle \cong \bigoplus_{(d,e) \in \mathbb{N}^2} T_{i,j}$ where each $T_{i,j}$ is a $\mathbf{k}$ -subspace which satisfy $T_{i,j}T_{i',j'} \subseteq T_{i+i',j+j'}$, for all $i, i', j, j' \in \mathbb{N}$. We can consider the grading $\mathbf{k}\langle X_1, \dots, X_n \rangle \cong \bigoplus_j (\bigoplus_i T_{i,j})$ in the second index only, and to say that an element f is e -homogenous means that $f \in \bigoplus_i T_{i,j}$ for some $j \in \mathbb{N}$. Note that since I is generated by e -homogenous elements, I is graded with $I = \bigoplus_j I_j$ where $I_j \subseteq \bigoplus_i T_{i,j}$ for all $j \in \mathbb{N}$. Moreover by the setup of our bigrading, where $\deg_{\mathcal{F}}(X_i) = d_i$, we have that $\mathcal{F}_n = \bigoplus_{i \leq n} \bigoplus_j T_{i,j}$.

Now, since $\pi : \mathbf{k}\langle X_1, \dots, X_n \rangle$ is a strict morphism, it follows that,

$$F_n = \pi(\mathcal{F}_n) = \pi \left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) = \frac{\bigoplus_{i \leq n} \bigoplus_j T_{i,j}}{\left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) \cap I}$$

for all $n \in \mathbb{N}$. Furthermore, since I is generated by e -homogenous elements by Lemma 59 we obtain a grading on $A = \bigoplus_j A_j$, where

$$A_j = \pi \left(\bigoplus_i T_{i,j} \right) = \frac{\bigoplus_i T_{i,j}}{\left(\bigoplus_i T_{i,j} \right) \cap I}.$$

Our first goal for this proof will be to show that $F_n = \bigoplus_j (F_n \cap A_j)$ for all $n \in \mathbb{N}$. Let us fix $n \in \mathbb{N}$, we observe that

$$F_n = \frac{\bigoplus_{i \leq n} \bigoplus_j T_{i,j}}{\left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) \cap I} \cong \frac{\left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) + I}{I} \cong \frac{\left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) + \bigoplus_j I_j}{\bigoplus_j I_j}$$

where we have used the Second Isomorphism Theorem for the first isomorphism and the grading on I for the second isomorphism. We can now reorder the sum in the numerator and use the grading in the second index j to write,

$$\frac{\left(\bigoplus_{i \leq n} \bigoplus_j T_{i,j} \right) + \bigoplus_j I_j}{\bigoplus_j I_j} \cong \frac{\bigoplus_j \left(\bigoplus_{i \leq n} T_{i,j} + I_j \right)}{\bigoplus_j I_j} \cong \bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j} + I_j}{I_j}$$

where the second isomorphism follows by repeated application of Lemma 62 and the observation that I_j is a submodule of $\bigoplus_{i \leq n} T_{i,j} + I_j$ for all j . Using the Second Isomorphism Theorem and the grading on I we may write,

$$\bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j} + I_j}{I_j} \cong \bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j}}{\left(\bigoplus_{i \leq n} T_{i,j} \right) \cap \left(\bigoplus_i T_{i,j} \cap I \right)} \cong \bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j}}{\left(\bigoplus_{i \leq n} T_{i,j} \right) \cap I}$$

where the second isomorphism follows since taking intersections is associative and $\bigoplus_{i \leq n} T_{i,j} \subseteq \bigoplus_i T_{i,j}$. We may apply the Second Isomorphism Theorem to the above,

together with the string of isomorphisms above, to obtain,

$$F_n \cong \bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j}}{\left(\bigoplus_{i \leq n} T_{i,j}\right) \cap I} \cong \bigoplus_j \frac{\bigoplus_{i \leq n} T_{i,j} + I}{I}. \quad (4.1)$$

Working from the other end, we see that after plugging in the expressions for F_n and A_j in terms of $T_{i,j}$ into $\bigoplus_{j \in \mathbb{N}} (F_n \cap A_j)$ we obtain

$$\bigoplus_j \left(\frac{\bigoplus_i T_{i,j}}{\bigoplus_i T_{i,j} \cap I} \cap \frac{\bigoplus_{i \leq n} \bigoplus_k T_{i,k}}{\bigoplus_{i \leq n} \bigoplus_k T_{i,k} \cap I} \right) \cong \bigoplus_j \left(\frac{\bigoplus_i T_{i,j} + I}{I} \cap \frac{\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I}{I} \right),$$

where the isomorphism follows by the Second Isomorphism Theorem. Now note that thinking of the free algebra as a left $\mathbf{k}$ -module and the modules $\bigoplus_i T_{i,j} + I$, $\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I$ and I as submodules we may invoke the Fourth Isomorphism Theorem, as well as the string of isomorphisms above, to write

$$\bigoplus_{j \in \mathbb{N}} (F_n \cap A_j) \cong \bigoplus_j \left(\frac{(\bigoplus_i T_{i,j} + I) \cap (\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I)}{I} \right). \quad (4.2)$$

Observe that the end expressions in Equations 4.1 and 4.2 look quite similar, and in fact if we can show that $\bigoplus_{i \leq n} T_{i,j} + I = (\bigoplus_i T_{i,j} + I) \cap (\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I)$ then our two ends will meet in the middle and it will follow that $F_n \cong \bigoplus_{j \in \mathbb{N}} (F_n \cap A_j)$. The inclusion $\bigoplus_{i \leq n} T_{i,j} + I \subseteq (\bigoplus_i T_{i,j} + I) \cap (\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I)$ is trivial. For the other inclusion, let $h \in (\bigoplus_i T_{i,j} + I) \cap (\bigoplus_{i \leq n} \bigoplus_k T_{i,k} + I)$, then on one hand we may write $h = \sum_i x_{i,j} + a$ where $x_{i,j} \in T_{i,j}$ and $a \in I$, whilst on the other hand we may write $h = \sum_{i \leq n} \sum_k y_{i,k} + b$ where $y_{i,k} \in T_{i,k}$ and $b \in I$. We use the grading on I to write $a = \sum_k a_k$ and $b = \sum_k b_k$, where $a_k, b_k \in I_k$. So far we have

$$\sum_i x_{i,j} + \sum_k a_k = \sum_{i \leq n} \sum_k y_{i,k} + \sum_k b_k = \sum_k \left(\sum_{i \leq n} y_{i,k} + b_k \right),$$

where we note that $\sum_{i \leq n} y_{i,k} + b_k$ is e -homogenous. Comparing the e -homogenous components of the above equality we see that for $k \neq j$ we have $a_k = \sum_{i \leq n} y_{i,k} + b_k$, and since $a_k \in I_k \subseteq I$ it follows that $\sum_{i \leq n} y_{i,k} + b_k \in I$, whilst for $k = j$ we have $\sum_i x_{i,j} + a_j = \sum_{i \leq n} y_{i,j} + b_j$, and since $\sum_i x_{i,j} \in \bigoplus_{i \leq n} T_{i,j}$ and $a_j \in I_j \subseteq I$ it follows that $\sum_{i \leq n} y_{i,j} + b_j \in \bigoplus_{i \leq n} T_{i,j} + I$. Therefore we may conclude that

$$h = \sum_{i \leq n} \sum_k y_{i,k} + b = \sum_{i \leq n} y_{i,j} + \left(b_j + \sum_{k \neq j} \sum_{i \leq n} y_{i,k} + b_k \right) \in \bigoplus_{i \leq n} T_{i,j} + I,$$

hence we obtain the other inclusion, and by our earlier remarks it now follows that $F_n = \bigoplus_j (F_n \cap A_j)$.

Now fix $n \in \mathbb{N}$, using $F_n = \bigoplus_j (F_n \cap A_j)$ we write

$$\frac{F_n}{F_{n-1}} \cong \frac{\bigoplus_j (A_j \cap F_n)}{\bigoplus_j (A_j \cap F_{n-1})} \cong \bigoplus_j \frac{A_j \cap F_n}{A_j \cap F_{n-1}},$$

where the second isomorphism holds by repeated use of Lemma 62. It now follows that

$$\mathrm{gr}_F A = \bigoplus_n \frac{F_n}{F_{n-1}} \cong \bigoplus_n \left(\bigoplus_j \frac{A_j \cap F_n}{A_j \cap F_{n-1}} \right) = \bigoplus_j \left(\bigoplus_n \frac{A_j \cap F_n}{A_j \cap F_{n-1}} \right),$$

hence defining $A'_{n,j} = (F_n \cap A_j) / (F_{n-1} \cap A_j)$ we see that we have $\mathrm{gr}_F A \cong \bigoplus_{(n,j) \in \mathbb{N}^2} A'_{n,j}$. Note that for any $n, n', j, j' \in \mathbb{N}$ we have

$$A'_{n,j} A'_{n',j'} = \frac{F_n \cap A_j}{F_{n-1} \cap A_j} \cdot \frac{F_{n'} \cap A_{j'}}{F_{n'-1} \cap A_{j'}} = \frac{F_{n+n'} \cap A_{j+j'}}{F_{n+n'-1} \cap A_{j+j'}} = A'_{n+n',j+j'},$$

where the middle equality follows by the definition of multiplication in $\mathrm{gr}_F A$ and the fact that $A = \bigoplus_j A_j$ is a grading. It now follows that $\mathrm{gr}_F A \cong \bigoplus_{(n,j) \in \mathbb{N}^2} A'_{n,j}$ is a bigrading. $\square$

Now that we have defined MS filtrations and discussed some of their nice properties, in the next chapter we will apply results developed in this chapter to comment on finite dimensional Noetherian filtrations on S .

Chapter 5

Progress on the MR Question

In this chapter we will attempt to make progress on the MR question with respect to the MS example S we have introduced in the last chapter. We have already seen how arbitrary finite dimensional filtrations can result in associated graded algebras with wildly different properties, and - in order to make progress - in this chapter we will solely focus on MS filtrations. Of course, the ideal scenario is one in which given an MS filtration $F_\bullet$ on S we could obtain a presentation for $\text{gr}_F S$ using which we could more easily gauge the Noetherianity of $\text{gr}_F S$. This is going to be focus of Section 5.1 below where for a certain type of MS filtrations we will compute a Groebner basis for the ideal generated by the relations in S using which we will find a presentation for $\text{gr}_F S$. In doing so, we will see the computational infeasibility of this approach, and we will give a different argument in Section 5.2 to show that for a certain class of MS filtrations the associated graded algebra is not Noetherian.

5.1 Finding the Presentation of the Associated Graded Algebra

Let $F_\bullet$ be an MS filtration on S , with the choice of x, y, z, w as the minimal generators of S and such that $\deg_F(x) = a$, $\deg_F(y) = b$, $\deg_F(z) = c$, $\deg_F(w) = d$ where $a, b, c, d \in \mathbb{Z}_+$. We will denote the associated graded algebra induced by this filtration by $\text{gr}^{(a,b,c,d)} S$.

5.1.1 Calculating a Groebner Basis

Here we aim to compute a Groebner basis for the ideal generated by the relations in S , in the process resolving all inclusion and overlap ambiguities which may arise. Specifically, we want to calculate the Groebner basis for the ideal I generated by,

$$\{xyz - yx, zw - 1, wz - zw, zx - xz, zy - yz, wx - xw, wy - yw\} \quad (5.1)$$

subject to the degree-lexicographic ordering with $x < y < z < w$. It is clear that the above set of relations is **not** ambiguity-free as for example the following overlap ambiguity $xyzw$ is not resolvable since on one hand,

$$xyzw = xy$$

is a reduced word and,

$$xyzw = yxw$$

is a reduced word, however these words are not equal with respect to our current relations. We claim that,

Lemma 64. *The following set,*

$$\begin{aligned} \mathcal{R} = \{ &zx - xz, zy - yz, wx - xw, wy - yw, zw - 1, wz - 1, \\ &xy^{n+1}x^mz - yxy^nx^m, yx^{n+1}y^mw - xyx^ny^m, \\ &yx^{n+2}y - yxy^nyx, yxy^nyx - xy^{n+2}x : n, m \in \mathbb{N} \} \end{aligned} \quad (5.2)$$

is a Groebner basis for the ideal generated by the relations in Equation 5.1, with respect to the degree-lexicographic ordering with $x < y < z < w$.

For the sake of clarity, let us label the above relations with numbers from 1 to 10 in order in which they appear (reading left-to-right, top-to-bottom) so that - for example - the relation $xy^{n+1}x^mz - yxy^nx^m$ has label 7. In the proof below the notation $=_i$ will be used to signify that a word has been reduced using the relation labelled i . We will sometimes signify repeated reductions using the same relation i with one occurrence of $=_i$. Furthermore, to save on space, we will sometimes use $=_{i,j}$ for $1 \leq i, j \leq 4$ to denote repeated applications of the relations labelled i and j , so that in one swift motion we may commute a z or a w past the x 's and y 's.

Proof (of Lemma 64). It should be clear that $\mathcal{R}$ generates I as an ideal and that all generators of $\mathcal{R}$ lie in I . Now we focus our attention on identifying and resolving any inclusion and overlap ambiguities. Unless otherwise stated, we assume that any exponent $n, m, l \in \mathbb{N}$. First note that we have one family of inclusion ambiguities, $yx^{n+2}y^mw$, where $m \geq 1$, which on one hand we can reduce,

$$yx^{n+2}y^mw =_9 xyx^nyxy^{m-1}w =_8 xyx^{n+1}y^m$$

whilst on the other hand we can reduce,

$$yx^{n+2}y^mw =_8 xyx^{n+1}y^m$$

hence the inclusion ambiguities are resolvable. Now we work through all overlap ambiguities, starting with $wyxy^nyx$,

$$\begin{aligned} wyxy^nyx &=_{3,4} yxy^nwxy =_8 xy^{n+1}xy \\ wyxy^nyx &=_{10} wxy^{n+2}x =_{3,4} xy^{n+2}xw =_8 xy^{n+1}xy \end{aligned}$$

Next $zyxy^nxy$, for $n \geq 1$ we have,

$$\begin{aligned} zyxy^nxy &=_{1,2} yxy^nxyz =_7 y^2xy^{n-1}xy =_{10} yxy^{n+1}x \\ zyxy^nxy &=_{10} zxy^{n+2}x =_{1,2} xy^{n+2}xz =_7 yxy^{n+1}x \end{aligned}$$

whilst for $n = 0$ we have,

$$\begin{aligned} zyx^2y &=_{1,2} yx^2yz =_{10} xy^2xz \\ zyx^2y &=_{10} zxy^2x =_{1,2} xy^2xz \end{aligned}$$

Next $yx^{n+2}yxy^mxy$, for $m \geq 1$ we have,

$$\begin{aligned} yx^{n+2}yxy^mxy &=_9 xyx^nyx^2y^mxy =_9 xyx^{n+1}y^2xy^{m-1}xy =_{10} xyx^{n+1}yxy^{m+1}x \\ yx^{n+2}yxy^mxy &=_{10} yx^{n+3}y^{m+2}x =_9 xyx^{n+1}yxy^{m+1}x \end{aligned}$$

whilst for $m = 0$ we have,

$$\begin{aligned} yx^{n+2}yx^2y &=_9 xyx^nyx^3y =_9 xyx^{n+1}yxyx \\ yx^{n+2}yx^2y &=_{10} yx^{n+3}y^2x =_9 xyx^{n+1}yxyx \end{aligned}$$

Next $yxy^mxyx^{n+2}y$, for $m \geq 1$,

$$\begin{aligned} yxy^mxyx^{n+2}y &=_{10} xy^{m+2}x^{n+3}y =_9 xy^{m+1}xyx^{n+1}yx \\ yxy^mxyx^{n+2}y &=_9 yxy^mx^2yx^nyx =_9 yxy^{m-1}xy^2x^{n+1}yx =_{10} xy^{m+1}xyx^{n+1}yx \end{aligned}$$

whilst for $m = 0$,

$$\begin{aligned} yx^2yx^{n+2}y &=_{10} xy^2x^{n+3}y =_9 xyxyx^{n+1}yx \\ yx^2yx^{n+2}y &=_9 yx^3yx^nyx =_9 xyxyx^{n+1}yx \end{aligned}$$

Next $yxy^nxyx^{m+1}y^lw$, for $n \geq 1$

$$\begin{aligned} yxy^nxyx^{m+1}y^lw &=_{10} xy^{n+2}x^{m+2}y^lw =_8 xy^{n+1}xyx^{m+1}y^l \\ yxy^nxyx^{m+1}y^lw &=_8 yxy^nx^2yx^my^l =_9 yxy^{n-1}xy^2x^{m+1}y^l =_{10} xy^{n+1}xyx^{m+1}y^l \end{aligned}$$

whilst for $n = 0$,

$$\begin{aligned} yx^2yx^{m+1}y^lw &=_{10} xy^2x^{m+2}y^lw =_8 xyxyx^{m+1}y^l \\ yx^2yx^{m+1}y^lw &=_8 yx^3yx^my^l =_9 xyxyx^{m+1}y^l \end{aligned}$$

Next $yxyxyw$,

$$\begin{aligned} yxyxyw &=_{10} xy^3xw =_8 xy^2xy \\ yxyxyw &=_8 yx^2y^2 =_9 xy^2xy \end{aligned}$$

Next yxy^nxyx^mz ,

$$\begin{aligned} yxy^nxyx^mz &=_{10} xy^{n+2}x^{m+1}z =_7 yxy^{n+1}x^{m+1} \\ yxy^nxyx^mz &=_7 yxy^{n+1}x^{m+1} \end{aligned}$$

Next $yx^{n+2}yx^{m+1}y^lw$,

$$\begin{aligned} yx^{n+2}yx^{m+1}y^lw &=_9 yxy^nyx^{m+2}y^lw =_8 yxy^{n+1}yx^{m+1}y^l \\ yx^{n+2}yx^{m+1}y^lw &=_8 yx^{n+3}yx^my^l =_9 yxy^{n+1}yx^{m+1}y^l \end{aligned}$$

Next $yx^{n+2}yx^mz$, for $n \geq 1$,

$$\begin{aligned} yx^{n+2}yx^mz &=_9 yxy^nyx^{m+1}z =_7 yxy^{n-1}yx^{m+2} \\ yx^{n+2}yx^mz &=_7 yxy^{n+1}yx^{m+1} =_9 yxy^{n-1}yx^{m+2} \end{aligned}$$

whilst for $n = 0$,

$$\begin{aligned} yx^2yx^mz &= {}_9 xy^2x^{m+1}z = {}_7 yxyx^{m+1} \\ yx^2yx^mz &= {}_7 yxyx^{m+1} \end{aligned}$$

Next $zyx^{n+2}y$,

$$\begin{aligned} zyx^{n+2}y &= {}_{1,2} yx^{n+2}yz = {}_9 yxy^n yxz \\ zyx^{n+2}y &= {}_9 zxyx^n yx = {}_{1,2} yxy^n yxz \end{aligned}$$

Next $wyx^{n+2}y$,

$$\begin{aligned} wyx^{n+2}y &= {}_{3,4} yx^{n+2}yw = {}_9 yxy^n yxw \\ wyx^{n+2}y &= {}_9 wxyx^n yx = {}_{3,4} yxy^n yxw \end{aligned}$$

Next $zyx^{n+1}y^mw$, for $n, m \geq 1$,

$$\begin{aligned} zyx^{n+1}y^mw &= {}_{1,2} yx^{n+1}y^mzw = {}_5 yx^{n+1}y^m = {}_9 yxy^{n-1}yxy^{m-1} \\ zyx^{n+1}y^mw &= {}_8 zxyx^n y^m = {}_{1,2} yxy^n y^mz = {}_7 yxy^{n-1}yxy^{m-1} \end{aligned}$$

whilst for $n = 0$ and $m \geq 0$,

$$\begin{aligned} zxyx^m w &= {}_{1,2} yxy^m zw = {}_5 yxy^m \\ zxyx^m w &= {}_8 zxy^{m+1} = {}_{1,2} xy^{m+1}z = {}_7 yxy^m \end{aligned}$$

whilst for $n \geq 0$ and $m = 0$,

$$\begin{aligned} zyx^{n+1}w &= {}_{1,2} yx^{n+1}zw = {}_5 yx^{n+1} \\ zyx^{n+1}w &= {}_8 zxyx^n = {}_{1,2} yxy^n z = {}_7 yx^{n+1} \end{aligned}$$

Next $wyx^{n+1}y^mw$,

$$\begin{aligned} wyx^{n+1}y^mw &= {}_{3,4} yx^{n+1}y^mw^2 = {}_8 yxy^n y^mw \\ wyx^{n+1}y^mw &= {}_8 wxyx^n y^m = {}_{3,4} yxy^n y^mw \end{aligned}$$

Next $yx^{n+1}y^mwz$, for $n, m \geq 1$,

$$\begin{aligned} yx^{n+1}y^mwz &= {}_8 yxy^n y^mz = {}_7 yxy^{n-1}yxy^{m-1} \\ yx^{n+1}y^mwz &= {}_6 yx^{n+1}y^m = {}_9 yxy^{n-1}yxy^{m-1} \end{aligned}$$

whilst for $n = 0$ and $m \geq 0$,

$$\begin{aligned} yxy^m wz &= {}_8 xy^{m+1}z = yxy^m \\ yxy^m wz &= {}_6 yxy^m \end{aligned}$$

whilst for $n \geq 0$ and $m = 0$,

$$\begin{aligned} yx^{n+1}wz &= {}_8 yxy^n z = {}_7 yx^{n+1} \\ yx^{n+1}wz &= {}_6 yx^{n+1} \end{aligned}$$

Next $yx^{n+1}y^mwx$, for $n \geq 1$ and $m \geq 2$,

$$\begin{aligned} yx^{n+1}y^mwx &= {}_8 yxy^n y^m x \\ yx^{n+1}y^mwx &= {}_3 yx^{n+1}y^m xw = {}_8 yx^{n+1}y^{m-1}xy = {}_9 yxy^{n-1}yxy^{m-2}xy = {}_{10} yxy^n y^m x \end{aligned}$$

whilst for $n = 0$ and $m \geq 1$,

$$\begin{aligned} yxy^m wx &= {}_8 xy^{m+1}x \\ yxy^m wx &= {}_3 yxy^m xw = {}_8 yxy^{m-1}xy = {}_{10} xy^{m+1}x \end{aligned}$$

whilst for $n = 0$ and $m = 0$,

$$\begin{aligned} yxwx &=_8 xyx \\ yxwx &=_{=3} yx^2w =_8 xyx \end{aligned}$$

whilst for $n \geq 1$ and $m = 1$,

$$\begin{aligned} yx^{n+1}ywx &=_8 xyx^n yx \\ yx^{n+1}ywx &=_{=3} yx^{n+1}yxw =_8 yx^{n+2}y =_9 xyx^n yx \end{aligned}$$

whilst for $n \geq 1$ and $m = 0$,

$$\begin{aligned} yx^{n+1}wx &=_8 xyx^{n+1} \\ yx^{n+1}wx &=_{=3} yx^{n+2}w =_8 xyx^{n+1} \end{aligned}$$

Next $yx^{n+1}y^mwy$,

$$\begin{aligned} yx^{n+1}y^mwy &=_8 xyx^n y^{m+1} \\ yx^{n+1}y^mwy &=_{=4} yx^{n+1}y^{m+1}w =_8 xyx^n y^{m+1} \end{aligned}$$

Next $wxy^{n+1}x^mz$,

$$\begin{aligned} wxy^{n+1}x^mz &=_{=3,4} xy^{n+1}x^m wz =_6 xy^{n+1}x^m \\ wxy^{n+1}x^mz &=_{=7} wxyx^n x^m =_{=3,4} yxwy^n x^m =_8 xy^{n+1}x^m \end{aligned}$$

Next $zxy^{n+1}x^mz$,

$$\begin{aligned} zxy^{n+1}x^mz &=_{=1,2} xy^{n+1}zx^m z =_7 yxy^n x^m z \\ zxy^{n+1}x^mz &=_{=7} zxyx^n x^m =_{=1,2} yxy^n x^m z \end{aligned}$$

Next $xy^{n+1}x^mzx$,

$$\begin{aligned} xy^{n+1}x^mzx &=_{=7} yxy^n x^{m+1} \\ xy^{n+1}x^mzx &=_{=1} xy^{n+1}x^{m+1}z =_7 yxy^n x^{m+1} \end{aligned}$$

Next $xy^{n+1}x^mzy$, for $n \geq 1$ and $m \geq 2$,

$$\begin{aligned} xy^{n+1}x^mzy &=_{=7} yxy^n x^m y =_9 yxy^{n-1}xyx^{m-2}yx =_{=10} xy^{n+1}x^{m-1}yx \\ xy^{n+1}x^mzy &=_{=2} xy^{n+1}x^m yz =_{=7} xy^{n+1}x^{m-1}yx \end{aligned}$$

whilst for $n = 0$ and $m \geq 1$,

$$\begin{aligned} xyx^mzy &=_{=7} yx^{m+1}y =_9 xyx^{m-1}yx \\ xyx^mzy &=_{=2} xyx^m yz =_{=7} xyx^{m-1}yx \end{aligned}$$

whilst for $n = 0$ and $m = 0$,

$$\begin{aligned} xyzzy &=_{=7} yxy \\ xyzzy &=_{=2} xy^2z =_{=7} yxy \end{aligned}$$

whilst for $n \geq 1$ and $m = 1$,

$$\begin{aligned} xy^{n+1}xzy &=_{=7} yxy^n xy =_{=10} xy^{n+2}x \\ xy^{n+1}xzy &=_{=2} xy^{n+1}xyz =_{=7} xy^{n+2}x \end{aligned}$$

whilst for $n \geq 1$ and $m = 0$,

$$\begin{aligned} xy^{n+1}zy &=_{=7} yxy^{n+1} \\ xy^{n+1}zy &=_{=2} xy^{n+2}z =_{=7} yxy^{n+1} \end{aligned}$$

Next $xy^{n+1}x^mzw$, for $n, m \geq 1$,

$$xy^{n+1}x^mzw =_7 yxy^n x^m w =_8 yxy^{n-1}xyx^{m-1} =_{10} xy^{n+1}x^m$$

$$xy^{n+1}x^mzw =_5 xy^{n+1}x^m$$

whilst for $n = 0$ and $m \geq 0$,

$$xyx^mzw =_7 yx^{m+1}w =_8 yx^m$$

$$xyx^mzw =_5 yx^m$$

whilst for $n \geq 0$ and $m = 0$,

$$xy^{n+1}zw =_7 yxy^n w =_8 xy^{n+1}$$

$$xy^{n+1}zw =_5 xy^{n+1}$$

Next $yx^{n+2}yw$,

$$yx^{n+2}yw =_8 yx^{n+1}y$$

$$yx^{n+2}yw =_9 yx^n yxw =_8 yx^{n+1}y$$

Next zwz ,

$$zwz =_5 z$$

$$zwz =_6 z$$

Next wzw ,

$$wzw =_6 w$$

$$wzw =_5 w$$

Next wzx ,

$$wzx =_6 x$$

$$wzx =_1 wxz =_3 xwz =_6 x$$

Next wzy ,

$$wzy =_6 y$$

$$wzy =_2 wyz =_4 ywz =_6 y$$

Next zwx ,

$$zwx =_5 x$$

$$zwx =_3 zxw =_1 xzw =_5 x$$

and finally zwy ,

$$zwy =_5 y$$

$$zwy =_4 zwy =_2 yzw =_5 y$$

□

5.1.2 Obtaining a Presentation for $\text{gr}^{(a,b,c,d)}S$

Onto the main result! The long calculation in Section 5.1.1 rewards us handsomely with an explicit basis $\mathcal{B}$ of S , which by the Diamond Lemma consists of images under the canonical surjection of reduced words in x, y, z, w . We may now proceed to give a presentation of the associated graded algebra $\text{gr}^{(a,b,c,d)}$.

Theorem 65. $\text{gr}^{(a,b,c,d)} \cong \mathbf{k}\langle X, Y, Z, W \rangle / K$ where K is the ideal generated by,

$$\{ZY - YZ, ZX - XZ, WY - YW, WX - XW, WZ, ZW, YXZ, XYW, \\ XY^nXY - XY^{n+2}X, YX^{n+2}Y - YX^{n+1}YX \mid n \in \mathbb{N}\}$$

Proof. By Theorem 58 it follows that $\text{gr}_F \mathcal{B}$ is a basis of $\text{gr}_F A$. Let $\pi' : \mathbf{k}\langle X_1, \dots, X_4 \rangle \rightarrow \mathbf{k}\langle X, Y, Z, W \rangle / K$ denote the canonical surjection and for an element $h \in \mathbf{k}\langle X_1, \dots, X_4 \rangle$ let $h(\underline{G}) := \pi'(h)$. Define $\mathcal{B}' := \pi'(\langle X \rangle_{\text{irr}})$. We claim that $\mathcal{B}'$ is a $\mathbf{k}$ -basis of $\mathbf{k}\langle X, Y, Z, W \rangle / K$. Let

$$\mathcal{R}' = \{ZX - XZ, ZY - YZ, WX - XW, WY - YW, ZW - 1, WZ - 1, \\ XY^{n+1}X^mZ, YX^{n+1}Y^mW, YX^{n+2}Y - YX^{n+1}YX, YXY^nXY - XY^{n+2}X\}, \quad (5.3)$$

and note that the ideal generated by $\mathcal{R}'$ is K . We go on to show that $\mathcal{R}'$ is a Groebner basis of K . We observe the relations in $\mathcal{R}'$ are closely related to the relations in $\mathcal{R}$ (see 5.2), up to relabelling, the relations in $\mathcal{R}'$ are the relations in $\mathcal{R}$ with the inhomogeneous parts of lesser degree (with respect to the deg-lex ordering) removed. Hence for every $r \in \mathcal{R}$, which we write $r = h(\underline{g}) - f(\underline{g})$ where $h(\underline{g})$ is the homogenous component of highest degree, we have the corresponding $r' \in \mathcal{R}'$ where $r' = h(\underline{G})$. Crucially, since the leading words in $\mathcal{R}'$ and $\mathcal{R}$ are the same, then the ambiguities present in $\mathcal{R}'$ are exactly those present in $\mathcal{R}$. Moreover, we can use the reductions in the proof in Section 5.1.1 to resolve the ambiguities in $\mathcal{R}'$, since the homogenous relations in $\mathcal{R}'$ and $\mathcal{R}$ are (again up to relabelling) the same and whenever we reduce by an inhomogeneous relation $r \in \mathcal{R}$ then the similar reduction with $r' \in \mathcal{R}'$ will reduce the word to 0. Since if one reduction-path contains an inhomogeneous relation the other reduction-path must contain an inhomogeneous relation, it follows that both reduction-paths are reduced to 0 in reducing the ambiguity in $\mathcal{R}'$. It follows that the resolvability of ambiguities in $\mathcal{R}$ gives the resolvability of ambiguities in $\mathcal{R}'$, hence by the Diamond Lemma it follows that $\mathcal{R}'$ is a Groebner basis and the words in X, Y, Z, W reduced with respect to the relations in $\mathcal{R}'$ form a $\mathbf{k}$ -basis of $\mathbf{k}\langle X, Y, Z, W \rangle / K$. The discussion above shows that if $\omega(\underline{g})$ is a reduced word in S (and hence in $\text{gr}^{(a,b,c,d)} S$) it follows that $\omega(\underline{G})$ is a reduced word in $\mathbf{k}\langle X, Y, Z, W \rangle / K$.

Now, we have a surjective morphism $\text{gr}\pi : \text{gr}_{\mathcal{F}} \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle \rightarrow \text{gr}^{(a,b,c,d)}$ and an isomorphism from Proposition 56, $\theta : \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle \rightarrow \text{gr}_{\mathcal{F}} \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle$ which compose to give a surjective morphism, $\text{gr}\pi \circ \theta : \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle \rightarrow \text{gr}^{(a,b,c,d)} S$, given by $(\text{gr}\pi \circ \theta)(X_i) = \text{gr}_F(g_i)$. We also consider the canonical surjection $\pi' : \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle \rightarrow \mathbf{k}\langle X, Y, Z, W \rangle / K$. Note that the generators of K are in the kernel $\ker(\text{gr}\pi \circ \theta)$, since the generators of K are precisely the relations in $\mathcal{R}$ modulo terms of lower degree. Thus we may invoke the universal property of factor algebras to conclude that we have a map $\hat{\theta} : \mathbf{k}\langle X, Y, Z, W \rangle / K \rightarrow \text{gr}^{(a,b,c,d)} S$ which makes the

following diagram commute,

$$\begin{array}{ccc} \mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle & \xrightarrow{\pi'} & \mathbf{k}\langle X, Y, Z, W \rangle / K \\ & \searrow \text{gr}\pi \circ \theta & \downarrow \hat{\theta} \\ & & \text{gr}^{(a,b,c,d)} S \end{array}$$

Since $\text{gr}\pi \circ \theta$ is surjective it follows that $\hat{\theta}$ is surjective. It now suffices to prove that $\hat{\theta}$ is injective. So let $\sum_i c_i \omega_i(\underline{G}) = 0$ where $\omega_i \in \langle X \rangle_{\text{irr}}$ and $c_i \in \mathbf{k}$ for all i , such that $\hat{\theta}(\sum_i c_i \omega_i(\underline{G})) = 0$. Our assumption, and the fact that $\hat{\theta}$ is an algebra morphism gives us,

$$0 = \hat{\theta} \left(\sum_i c_i \omega_i(\underline{G}) \right) = \sum_i c_i \omega_i(\hat{\theta}(\underline{G})) = \sum_i c_i \omega_i(\text{gr}(\underline{g})).$$

For all i we have that $\omega_i(\text{gr}_F(\underline{g})) \in \text{gr}_F B$ which is a basis of $\text{gr}_F A$, so it follows that $c_i = 0$ for all i . This implies that $\sum_i c_i \omega_i(\underline{G}) = 0$ is the trivial linear combination and thus manifestly $\hat{\theta}$ is injective and we are done. $\square$

5.2 Non-Noetherianity of Monomial MS Filtrations

Having established a negative result of the conjecture for a special class of MS filtrations, we aim to generalise our result with the goal of this subsection being the following result,

Theorem 66. *Let $F_\bullet$ be a monomial MS filtration on S , then $\text{gr}_F S$ is not Noetherian.*

which we will obtain by imitating the proof in Section 3.5 of [MS89] which shows that $\text{gr}^{(1,1,1,d)} S$ is not Noetherian. First we will show some lemmas.

Lemma 67. *S is a $\mathbb{N}$ -bigraded algebra in degrees of x and y .*

Proof. We have a grading $\mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle = \bigoplus_j T_j$ on the free algebra defined by setting $\deg(X_1) = 1$ and $\deg(X_i) = 0$ for $i \in \{2, 3, 4\}$. Let x, y, z, w be the minimal generating set of S , such that $\pi(X_1) = x$, then noting that all relations S are homogenous in the degree of x it follows that $\ker(\pi)$ is generated by homogenous generators with respect to the grading T_j . It now follows by Lemma 59 that $S = \bigoplus_j \pi(T_j)$ is graded, where $\pi(T_j)$ will be precisely the $\mathbf{k}$ -span of words in x, y, z, w with j occurrences of x . Defining $T'_j = \pi(T_j)$, we obtain the first half of the result, i.e. S is $\mathbb{N}$ -graded in degree of x . The second half of the result follows analogously by considering the grading $\mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle = \bigoplus_i R_i$ defined by setting $\deg(X_2) = 1$ and $\deg(X_i) = 0$ for $i \in \{1, 3, 4\}$. $\square$

For the next result we recall that the Diamond Lemma gives us a preferred basis $\mathcal{B}$ of S which behaves nicely with respect to the map gr_F . For results about $\mathcal{B}$ please see Section 4.3.

Lemma 68. $\text{gr}^{(a,b,c,d)}S$ is a $\mathbb{N}$ -bigraded algebra in degrees of $\bar{x}$ and $\bar{y}$.

Proof. Take the grading T_j on the free algebra $\mathbf{k}\langle X_1, X_2, X_3, X_4 \rangle$ as defined in Lemma 67, and recall that $\ker \varphi$ is generated by homogenous elements with respect to T_j , where we note that we let $\{x, y, z, w\}$ be the minimal generating set of S . It now follows by Lemma 63 that $\text{gr}^{(a,b,c,d)}S = \bigoplus_j S_j$ is a grading of the associative graded algebra where $S_j = \bigoplus_n (F_n \cap T'_j) / (F_{n-1} \cap T'_j)$ where note $T'_j = \pi(T_j)$.

Now recall that the Diamond Lemma gives us a basis $\mathcal{B}$ of S , which we can use to express any monomial in S , hence in the definition of the grading T_j we may assume without loss of generality that $\omega(\underline{g}) \in \mathcal{B}$, where $\omega \in \langle X \rangle_{\text{irr}}$. Thus observe that for any $n, j \in \mathbb{N}$, $F_n \cap T'_j$ is precisely the $\mathbf{k}$ -span of words of F -degree n with j occurrences of x . It follows by Lemma 51 that for $\omega(\underline{g}) \in F_n \cap T'_j$ we have $\deg_{\mathcal{F}}(\omega) = \deg_F \omega(\underline{g})$, and hence by Lemma 52 it follows that $\text{gr}_F(\omega(\underline{g})) = \omega(\text{gr}_F(\underline{g}))$. It now follows that, $(F_n \cap T'_j) / (F_{n-1} \cap T'_j) = \text{gr}_F(F_n \cap T'_j)$ is the $\mathbf{k}$ -span of $\text{gr}_F(\omega(\underline{g})) = \omega(\text{gr}_F(\underline{g}))$, i.e. words in F_n / F_{n-1} with j occurrences of $\text{gr}_F(x) = \bar{x}$.

It follows by the above that $S_j = \bigoplus_n (F_n \cap T'_j) / (F_{n-1} \cap T'_j)$ is the $\mathbf{k}$ -span of all words in $\bar{x}, \bar{y}, \bar{z}, \bar{w}$ which have exactly j occurrences of $\bar{x}$, hence $\text{gr}^{(a,b,c,d)}S$ is graded in degree of $\bar{x}$. The argument showing that $\text{gr}_F S$ is graded in degree of $\bar{y}$ follows mutatis mutandis from the argument above. $\square$

Lemma 69. $\bar{z}, \bar{w}$ are central elements of $\text{gr}_F^{(a,b,c,d)}S$.

Proof. The following relations in S ,

$$zx - xz, \quad zy - yz, \quad wx - xw, \quad wy - yw, \quad wz - zw,$$

are homogenous in F -degree. It follows by the definition of multiplication in $\text{gr}_F^{(a,b,c,d)}S$ that in $\text{gr}_F^{(a,b,c,d)}S$ we have the relations,

$$\bar{z}\bar{x} - \bar{x}\bar{z}, \quad \bar{z}\bar{y} - \bar{y}\bar{z}, \quad \bar{w}\bar{x} - \bar{x}\bar{w}, \quad \bar{w}\bar{y} - \bar{y}\bar{w}, \quad \bar{w}\bar{z} - \bar{z}\bar{w},$$

and since $\text{gr}_F^{(a,b,c,d)}S$ is generated by $\bar{x}, \bar{y}, \bar{z}, \bar{w}$, together with the above relations it follows that $\bar{z}, \bar{w}$ are central. $\square$

We now present and expand upon the McConnell-Stafford proof below.

Proposition 70 ([MS89], Section 3.5). $\text{gr}^{(1,1,1,d)}S$ is not right Noetherian.

Proof. We observe that since in S we have $zxy = xyz = yx \in F_2$ it follows that in $\text{gr}^{(1,1,1,d)}S$ we have,

$$(\bar{z}\bar{x})\bar{y} = zxy + F_2 = 0 + F_2 = 0$$

We also note that for $m \in \mathbb{Z}_+$ we have $\deg_F(y^m(zx)) = m + 2$ since we can't reduce $y^m zx$ with any relation in S , thus we have $\bar{y}^m \bar{z} \bar{x} \neq 0$. We now define right ideals, $I_m = \bar{y}^m(\bar{z}\bar{x})\text{gr}^{(1,1,1,d)}S$ for each $m \geq 1$, and consider the right ideal $I = \sum_{m \geq 1} I_m$. We claim that I is not finitely generated, which by definition implies that $\text{gr}^{(1,1,1,d)}S$ is not right Noetherian.

By Lemma 68 we have a grading $\text{gr}^{(1,1,1,d)}S = \bigoplus_{i \in \mathbb{N}} \bar{R}_i$ in the degree of $\bar{y}$, and note that we have shown above that $\bar{z}\bar{x}\bar{y} = 0$. Both of these facts together with the observation that the generators of I_m are homogenous with respect to the grading in the degree of $\bar{y}$ let us deduce that $\forall m \in \mathbb{Z}_+$, $I_m \subseteq \bar{R}_m$. This allows us to upgrade the sum in the definition of I into a direct sum, i.e. we may write $I = \bigoplus_{m=1}^{\infty} I_m$.

Now, let $f_1, \dots, f_u \in I$ for some $u \in \mathbb{N}$, and define the right ideal $J := \sum_{j=1}^u f_j \text{gr}^{(1,1,1,d)}S$. We will show that $J \neq I$, by showing that $\exists M \in \mathbb{N}$ such that $J \subseteq \bigoplus_{m=1}^M I_m$. For each $j = 1, \dots, u$ we write $f_j = \sum_{l=1}^{M_j} h_{jl}$ where $h_{jl} \in I_l$ and $M_j \in \mathbb{N}$. Since there is a finite number of f_j we may define $M := \max\{M_j : 1 \leq j \leq u\}$. By construction of M and the grading of $\text{gr}^{(1,1,1,d)}S$ in the degree of $\bar{y}$, it follows that $\forall s_1, \dots, s_u \in \text{gr}^{(1,1,1,d)}S$, $\sum_{j=1}^u f_j s_j = \sum_{j=1}^u \sum_{l=1}^{M_j} h_{jl} s_j \in \bigoplus_{m=1}^M I_m$, since $h_{jl} s_j \in I_l$ for all j, l as $h_{jl} \in I_l$ and I_l is an ideal. This implies that $J \subseteq \bigoplus_{m=1}^M I_m \subset I$, where the strict inclusion follows since $\bar{y}^{M+1} \bar{z} \bar{x} \in I$ as by definition of I we have $I_{M+1} \subseteq I$, but $\bar{y}^{M+1} \bar{z} \bar{x} \notin J$ as $\bar{y}^{M+1} \bar{z} \bar{x} \notin \bigoplus_{m=1}^M I_m$. Therefore, if I were to be finitely generated then we could show that the finite collection of generators does not generate I , a clear contradiction, hence it follows that I is not finitely generated. The result now follows. $\square$

Having reviewed the McConnell-Stafford proof, we proceed to generalise it for a wider class of filtrations. To do this we would ideally conceive a classification of all minimal generating sets of S , however classifying the minimal generating sets of S is a **hard** problem and in this section we will satisfy ourselves by focusing only on a special class of generating sets which we define below.

Definition 71. A generating set G of S is called a monomial generating set if $\forall g \in G$, g is a monomial in x, y, z, w .

We now proceed to give a classification of monomial generating sets of S where recall that we are assuming in this paper that S is minimally generated by 4 elements.

Proposition 72. On S , all monomial generating sets have one of the following forms, where $\rho, \sigma, \tau, v \in \mathbb{N}$, $\tau, v \geq 1$ and $\gcd(\tau, v) = 1$,

1. $\{xz^\rho, yz^\sigma, z^\tau, w^v\}$ where additionally $\rho \geq 0, \sigma \geq 0$,
2. $\{xz^\rho, yw^\sigma, z^\tau, w^v\}$ where additionally $\rho \geq 0, \sigma \geq 1$,
3. $\{xw^\rho, yz^\sigma, z^\tau, w^v\}$ where additionally $\rho \geq 1, \sigma \geq 0$,
4. $\{xw^\rho, yw^\sigma, z^\tau, w^v\}$ where additionally $a \geq 1, b \geq 1$.

Proof. Let's denote the monomial generating set by $G = \{g_1, g_2, g_3, g_4\}$. By Lemma 67 we have a grading on S in degrees of x and y , and since G generates x - as G generates S by assumption - it follows that some g_i must be linear in x and contain no powers of y , i.e. $g_i = xz^{\rho_1}w^{\sigma_1}$ for some $\rho_1, \sigma_1 \in \mathbb{N}$. Note that we have used the fact that z, w are central in S to write the monomial in this form. By a similar argument it follows that for some g_j , where $i \neq j$, we have $g_j = yz^{\rho_2}w^{\sigma_2}$, for some $\rho_2, \sigma_2 \in \mathbb{N}$. Since we can reshuffle the indices as we please, without loss of generality we may say that $g_1 = xz^{\rho_1}w^{\sigma_1}$ and $g_2 = yz^{\rho_2}w^{\sigma_2}$. By again appealing to the grading on S , we note that for G to generate z we must have some g_k which does not contain any powers of x nor y , hence $g_k = z^{\rho_3}w^{\sigma_3}$ for $\rho_3, \sigma_3 \in \mathbb{N}$ such that they're not both zero. Note that if $\rho_3, \sigma_3 \in \mathbb{Z}_+$ we could reduce this with the relation $zw - 1$, hence it follows that after performing the necessary reductions we have $g_k = z^{\tau_3}$ or $g_k = w^{v_3}$ for some $\tau_3, v_3 \in \mathbb{Z}_+$. Again without loss of generality say $g_k = g_3$. If $g_3 = z^{\tau_3}$, then to generate w - again appealing to the grading in S - we require that $g_4 = w^{v_4}$ for some $v_4 \in \mathbb{Z}_+$, such that $\gcd(\tau_3, v_4) = 1$. Note that if τ_3, v_4 were not coprime then we could only generate powers of z which are multiples of $\gcd(\tau_3, v_4)$ using g_3, g_4 and the relation $zw - 1$. In a similar fashion one argues that if $g_3 = w^{v_3}$ then $g_4 = z^{\tau_4}$ where $\gcd(v_3, \tau_4) = 1$. By reshuffling indices, we may without loss of generality set $g_3 = z^\tau$ and $g_4 = w^v$ for some $\tau, v \in \mathbb{Z}_+$ such that $\gcd(\tau, v) = 1$. We thus obtain the monomial generating set

$$\{xz^{\rho_1}w^{\sigma_1}, yz^{\rho_2}w^{\sigma_2}, z^\tau, w^v\}.$$

Note that by using the relation $zw - 1$ we may reduce g_1 and g_2 , and depending on $\rho_1, \sigma_1, \rho_2, \sigma_2$ the above generating will be reduced to one of the forms (1)–(4). The additional constraints on ρ and σ are only present to prevent double counting of monomial generating sets. $\square$

Remark 73. Note that for each form of minimal monomial generating set G in the proposition above, any element of G is of the form $\omega(\underline{g})$ where $\omega \in \langle X \rangle_{\text{irr}}$.

We now consider a monomial generating set of S of the first form in the proposition above. The following argument will apply *mutatis mutandis* to the monomial generating sets of other forms.

Relabelling the generators as,

$$\alpha := xz^\rho, \quad \beta := yz^\sigma, \quad \gamma := z^\tau, \quad \delta := w^v,$$

we now strive to obtain relations for the old generators x, y, z, w in terms of the new generators $\alpha, \beta, \gamma, \delta$, and to rewrite the relations in S in terms of the new generators. Since $\gcd(\tau, v) = 1$, by Bézout's Lemma it follows that $\exists \lambda_1, \mu_1 \in \mathbb{Z}$ such that,

$$1 = \tau\lambda_1 + (-v)\mu_1 = \tau\lambda_1 - v\mu_1 \tag{5.4}$$

and similarly $\exists \lambda_2, \mu_2 \in \mathbb{Z}$ such that,

$$1 = (-\tau)\lambda_2 + v\mu_2 = -\tau\lambda_2 + v\mu_2 \quad (5.5)$$

These are linear equations in τ and v , and in fact they define two parallel lines in $\mathbb{R}^2$ with gradient τ/v . Despite the fact that there are an infinite number of integer solutions to the linear equations above, by imposing the restriction that $\lambda_1, \mu_1, \lambda_2, \mu_2 \in \mathbb{N}$, and by asking for solutions which are minimal with respect to the standard ordering on $\mathbb{N}$, we in fact obtain a unique solution! This can be further justified by noting that since the gradient of these lines is positive $\tau/v > 0$, as by assumption $\tau, v \geq 1$, there exist positive solutions to Equations 5.4 & 5.5 as given a solution (λ_1, μ_1) to Equation 5.4 we see that for all $n \in \mathbb{N}$,

$$\tau(\lambda_1 + nv) - v(\mu_1 + n\tau) = \tau\lambda_1 + \tau nv - v\mu_1 + vn\tau = \tau\lambda_1 - v\mu_1 = 1$$

hence we can pick $n \in \mathbb{N}$ such that $\lambda_1 + nv \geq 0$ and $\mu_1 + n\tau \geq 0$, for example we could pick $n = \lceil \max\{-\lambda_1/v, -\mu_1/\tau, 0\} \rceil$. We pick the solution (λ_1, μ_1) such that $\lambda_1 \in \mathbb{N}$ is minimal with respect to the standard ordering on $\mathbb{N}$, and since Equation 5.4 is linear this means that there is a unique solution with such minimal value λ_1 . In an analogous manner we obtain the unique solution (λ_2, μ_2) for Equation 5.5. With these solutions, note that we obtain the following expressions for z and w ,

$$\begin{aligned} z &= z^{\tau\lambda_1 - v\mu_1} = z^{\tau\lambda_1} w^{v\mu_1} = \gamma^{\lambda_1} \delta^{\mu_1}, \\ w &= w^{-\tau\lambda_2 + v\mu_2} = z^{\tau\lambda_2} w^{v\mu_2} = \gamma^{\lambda_2} \delta^{\mu_2}, \end{aligned}$$

and note that by construction these expressions are minimal in the sense that if $z = \gamma^\lambda \delta^\mu$ then $\lambda_1 \leq \lambda$ and $\mu_1 \leq \mu$, and in an analogous fashion the expression for w is minimal. Using the above, we can obtain expressions for x and y as follows,

$$\begin{aligned} x &= xz^\rho w^\rho = \alpha(\gamma^{\lambda_2} \delta^{\mu_2})^\rho = \alpha\gamma^{\rho\lambda_2} \delta^{\rho\mu_2}, \\ y &= yz^\sigma w^\sigma = \beta(\gamma^{\lambda_2} \delta^{\mu_2})^\sigma = \beta\gamma^{\sigma\lambda_2} \delta^{\sigma\mu_2}, \end{aligned}$$

and using these expressions we may rewrite the relations in S in terms of the new generators. After simplifying, the relation $xyz - yx$ becomes,

$$\alpha\beta\gamma^{\rho\lambda_2 + \sigma\lambda_2 + \lambda_1} \delta^{\rho\mu_2 + \sigma\mu_2 + \mu_1} - \alpha\beta\gamma^{\rho\lambda_2 + \sigma\lambda_2} \delta^{\rho\mu_2 + \sigma\mu_2} \quad (5.6)$$

and the relation $zw - 1$ becomes

$$\gamma^{\lambda_1 + \lambda_2} \delta^{\mu_1 + \mu_2} - 1$$

Our choice of $\lambda_1, \lambda_2, \mu_1, \mu_2$ allows us to deduce an important relationship captured in the lemma below,

Lemma 74. $\lambda_1 + \lambda_2 = v$ and $\mu_1 + \mu_2 = \tau$.

Proof. Subtracting Equation 5.4 from Equation 5.5 we obtain,

$$0 = -(\lambda_1 + \lambda_2)\tau + (\mu_1 + \mu_2)v \implies (\lambda_1 + \lambda_2)\tau = (\mu_1 + \mu_2)v$$

By construction $\lambda_1, \lambda_2, \mu_1, \mu_2 \in \mathbb{N}$, and since $(0, 0)$ is not a solution to Equations 5.4 and 5.5 it follows that $\lambda_1 + \lambda_2, \mu_1 + \mu_2 \in \mathbb{Z}_+$. Since τ and v are coprime, $(\lambda_1 + \lambda_2)\tau = (\mu_1 + \mu_2)v$ implies that $v \mid (\lambda_1 + \lambda_2)$ and $\tau \mid (\mu_1 + \mu_2)$, hence it follows that $\lambda_1 + \lambda_2 = \kappa v$ and $\mu_1 + \mu_2 = \kappa'\tau$ for some $\kappa, \kappa' \in \mathbb{Z}_+$. Note that this along with Equation 5.4 implies,

$$1 = \tau(\kappa v - \lambda_2) - v(\kappa'\tau - \mu_2) = (\kappa - \kappa')\tau v - \lambda_2\tau + \mu_2v$$

and using Equation 5.5 we see that the above becomes,

$$0 = (\kappa - \kappa')\tau v \implies \kappa = \kappa'$$

since $\tau v \neq 0$. It now suffices to prove that $\kappa = 1$. Assume for a contradiction that $\kappa \geq 2$, then either $\lambda_1 \geq v$ or $\lambda_1 < v$. If $\lambda_1 < v$ then $\lambda_2 \geq v$ and using Equation 5.5 we deduce that

$$\mu_2 = \frac{\tau\lambda_2 + 1}{v} \geq \frac{\tau v + 1}{v} = \tau + \frac{1}{v} > \tau$$

where the last inequality follows since $v \geq 1$. We observe that the above implies $\lambda_2 - v \geq 0$ and $\mu_2 - \tau > 0$. Moreover

$$-\tau(\lambda_2 - v) + v(\mu_2 - \tau) = -\tau\lambda_2 + \tau v + v\mu_2 - v\tau = 1,$$

thus $(\lambda_2 - v, \mu_2 - \tau)$ is a non-negative solution to Equation 5.5 with $\lambda_2 - v < \lambda_2$, but this contradicts our minimal choice of (λ_2, μ_2) .

Now we assume that $\lambda_1 \geq v$, and using Equation 5.4 we deduce that

$$\mu_1 = \frac{\lambda_1\tau - 1}{v} \geq \frac{v\tau - 1}{v} = \tau - \frac{1}{v}.$$

If $v > 1$, then $\tau - 1/v > \tau - 1$ and since $\mu_1 \in \mathbb{N}$ it follows from the above that in this case $\mu_1 \geq \tau$. Now observe that $\lambda_1 - v \geq 0$, $\mu_1 - \tau \geq 0$, and

$$(\lambda_1 - v)\tau - (\mu_1 - \tau)v = \lambda_1\tau - v\tau - \mu_1v + \tau v = 1,$$

hence $(\lambda_1 - v, \mu_1 - \tau)$ is a non-negative solution to Equation 5.4 with $\lambda_1 - v < \lambda_1$, but this contradicts our minimal choice of (λ_1, μ_1) .

If $v = 1$, then $\mu_1 \geq \tau - 1$, and by the argument above we see that to avoid a contradiction we require $\mu_1 = \tau - 1$, which forces $\mu_2 \geq \tau$ by our assumption that $\kappa \geq 2$. Note that

$\mu_1 = \tau - 1$ together with Equation 5.4 implies that

$$\lambda_1 = \frac{\mu_1 + 1}{\tau} = \frac{\tau - 1 + 1}{\tau} = 1,$$

and note that using $\lambda_1 + \lambda_2 = \kappa$ we deduce that $\lambda_2 = \kappa - 1 \geq 1$ since by assumption $\kappa \geq 2$. Observe that $\lambda_2 - 1 \geq 0$, $\mu_2 - \tau \geq 0$ and

$$-\tau(\lambda_2 - 1) + (\mu_2 - \tau) = -\tau\lambda_2 + \tau + \mu_2 - \tau = 1,$$

hence $(\lambda_2 - 1, \mu_2 - \tau)$ is a non-negative solution to Equation 5.5 with $\lambda_2 - 1 < \lambda_2$, but this contradicts our minimal choice of (λ_2, μ_2) .

We now see that our assumption that $\kappa \geq 2$ leads to a contradiction, thus it follows that $\kappa = 1$ and the result follows. $\square$

Using Lemma 74 it follows that the relation $zw - 1$ becomes

$$\gamma^v \delta^\tau - 1, \tag{5.7}$$

which is reduced since $\text{lcm}(\tau, v) = \tau v$ as $\text{gcd}(\tau, v) = 1$. However, the first relation is probably not reduced, and we could further reduce it with the second relation. Before proceeding to find a reduced form of relation 5.6 we prove lemmas which will be needed to recreate the McConnell-Stafford proof.

Remark 75. Note that all relations in S are homogenous with respect to degrees of α and β . We have already seen how the analogues of relations $xyz - yx$ and $zw - 1$ are homogenous in degree of α and β ; in fact this also holds for the other relations which capture the fact that z and w are central. Taking $zx - xz$ as an example we see that the analogous relation in the new generators is,

$$\gamma\alpha = z^\tau(xz^\rho) = xz^{\rho+\tau} = (xz^\rho)z^\tau = \alpha\gamma$$

where we have used the relation $zx - xz$ to commute z^τ past x . In a similar fashion, leveraging the appropriate relations when needed, we observe that the remaining relations in S written in the new generators become $\gamma\beta = \beta\gamma$, $\delta\alpha = \alpha\delta$, $\delta\beta = \beta\delta$, $\gamma\delta = \delta\gamma$, hence all relations preserve the degree of α and β . Thus all relations in S preserve the degree of α and β .

Lemma 76. *The following statements hold:*

1. S is $\mathbb{N}$ -bigraded in the degrees of α and β .
2. $\text{gr}_F(S)$ is $\mathbb{N}$ -bigraded in the degrees of α and β .
3. $\bar{\gamma}, \bar{\delta}$ are central elements of $\text{gr}_F S$.

Proof. For (1), observe that all relations in S are homogenous with respect to the degrees of α and β , hence this will follow mutatis mutandis by the proof of Lemma 67. For (2), this follows mutatis mutandis from the proof of Lemma 68. For (3), note that by Remark 75 it follows that γ, δ commute with $\alpha, \beta, \gamma, \delta$, and since $\alpha, \beta, \gamma, \delta$ generate S it follows that γ, δ are central elements of S . We can now apply mutatis mutandis the proof of Lemma 69 to conclude that $\bar{\gamma}, \bar{\delta}$ are central elements of $\text{gr}_F S$. $\square$

Now we will find the reduced form of relation 5.6. Begin by writing $\lambda_2(\rho + \sigma) = q_1 v + r_1$ and $\mu_2(\rho + \sigma) = q_2 \tau + r_2$ where $0 \leq r_1 < d$, $0 \leq r_2 < c$ and $q_1, q_2 \in \mathbb{N}$. Then we can perform $q = \min\{q_1, q_2\}$ reductions using relation 5.7, and note that since $\lambda_2(\rho + \sigma) + \lambda_1 \geq \lambda_2(\rho + \sigma)$ and $\mu_2(\rho + \sigma) + \mu_1 \geq \mu_2(\rho + \sigma)$ it follows that we can perform the same number of reductions on the left hand side of relation 5.6. Whilst the right hand side is reduced now, the left hand side may not be, i.e. it may be the case that

$$\lambda_2(\rho + \sigma) + \lambda_1 - qv \geq v \quad \text{and} \quad \mu_2(\rho + \sigma) + \mu_1 - q\tau \geq \tau,$$

in which case we can perform additional reductions, say in total t of them, such that the left hand side is reduced. After performing the additional reductions, we are left with

$$\alpha\beta\gamma^{\lambda_2(\rho+\sigma)+\lambda_1-(q+t)v}\delta^{\mu_2(\rho+\sigma)+\mu_1-(q+t)\tau} - \beta\alpha\gamma^{\lambda_2(\rho+\sigma)-qv}\delta^{\mu_2(\rho+\sigma)-q\tau}.$$

We now wish to show that the F -degree of the left hand side word is strictly lower than the F -degree of the right hand side word in the expression above. Let $d_g := \deg_F(g)$ for $g \in \{\alpha, \beta, \gamma, \delta\}$, since both words are now reduced by assumption it suffices to check that

$$d_\alpha + d_\beta + d_\gamma(\lambda_2(\rho + \sigma) + \lambda_1 - (q + t)v) + d_\delta(\mu_2(\rho + \sigma) + \mu_1 - (q + t)\tau)$$

is strictly less than

$$d_\alpha + d_\beta + d_\gamma(\lambda_2(\rho + \sigma) - qv) + d_\delta(\mu_2(\rho + \sigma) - q\tau),$$

which after simplifying is equivalent to,

$$d_\gamma(\lambda_1 - tv) + d_\delta(\mu_1 - t\tau) < 0. \quad (5.8)$$

By definition of MS filtrations, $d_\gamma, d_\delta > 0$. Observe that using Lemma 74 we can write, $\lambda_1 - tv = \lambda_1(1 - t) - \lambda_2 t$ and $\mu_1 - t\tau = \mu_1(1 - t) - \mu_2 t$. Since by assumption $\lambda_1, \lambda_2, \mu_1, \mu_2 \in \mathbb{N}$ and $t \in \mathbb{Z}_+$ it follows from the rewritten expressions that $\lambda_1 - tv \leq 0$ and $\mu_1 - t\tau \leq 0$. Furthermore, by using Equation 5.5 we notice that

$$\mu_2 = \frac{1 + \lambda_2 \tau}{v} > 0$$

since $1 + \nu_2 c > 0$ and $d > 0$; we deduce that $t\mu_2 > 0$, hence we obtain a strict inequality $\mu_1 - t\tau = \mu_1(1 - t) - \mu_2 t < 0$. It now follows that $d_\gamma(\lambda_1 - tv) \leq 0$ and $d_\delta(\mu_1 - t\tau) < 0$,

hence Inequality 5.8 holds.

To recap, when $t = 0$ (i.e. we can perform no more reductions to the left hand side than to the right hand side) it is clear that in relation 5.6 the left hand side word has strictly higher F -degree than the right hand side word. Moreover, when $t \geq 1$ (i.e we can perform additional reductions to the left hand side word), by the work above it follows that the right hand side word has strictly higher F -degree than the left hand side word.

Remark 77. We have shown that the relation $xyz - yx$ becomes $\alpha\beta\gamma^{\eta_1}\delta^{\eta_2} - \beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2}$ for some $\eta_1, \eta_2, \zeta_1, \zeta_2 \in \mathbb{N}$ where $\deg_F(\alpha\beta\gamma^{\eta_1}\delta^{\eta_2}) \neq \deg_F(\beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2})$, for a generating set of the form (1) in Proposition 72. The upshot is that we can perform a similar analysis for generating sets of the form (2),(3),(4) in Proposition 72 and obtain the same result! We capture this in the following lemma.

Lemma 78. *Fix a monomial generating set of S , then $\alpha\beta\gamma^{\eta_1}\delta^{\eta_2} - \beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2}$ is a relation with $\alpha\beta\gamma^{\eta_1}\delta^{\eta_2}$ and $\beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2}$ both reduced words, such that $\deg_F(\alpha\beta\gamma^{\eta_1}\delta^{\eta_2}) \neq \deg_F(\beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2})$ for some $\eta_1, \eta_2, \zeta_1, \zeta_2 \in \mathbb{N}$.*

Proof. If the monomial generating set of S is of the form (1) in Proposition 72 then the work above establishes the result for this case. The proof for the other cases, that is for monomial generating sets of the form (2), (3) and (4) in Proposition 72 follows mutatis mutandis from the proof for case (1). $\square$

We are at last in a position to generalise the McConnell-Stafford proof and prove Theorem 66

Proof. (of Theorem 66). By Lemma 78 we have the relation $\alpha\beta\gamma^{\eta_1}\delta^{\eta_2} - \beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2}$, where $\deg_F(\alpha\beta\gamma^{\eta_1}\delta^{\eta_2}) \neq \deg_F(\beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2})$. Let us label $N_L = \deg_F(\alpha\beta\gamma^{\eta_1}\delta^{\eta_2})$ and $N_R = \deg_F(\beta\alpha\gamma^{\zeta_1}\delta^{\zeta_2})$, for now we assume that $N_L > N_R$. We will comment on the $N_L < N_R$ case later.

We observe that in $\text{gr}_F S$ the above relation in S implies

$$(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2})\bar{\beta} = \bar{\alpha}\bar{\beta}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2} = \alpha\beta\gamma^{\eta_1}\delta^{\eta_2} + F_{N_L-1} = \alpha\beta\gamma^{\zeta_1}\delta^{\zeta_2} + F_{N_L-1} = 0,$$

since by assumption $N_L > N_R$ and $\alpha\beta\gamma^{\zeta_1}\delta^{\zeta_2} \in F_{N_R}$ hence $\alpha\beta\gamma^{\zeta_1}\delta^{\zeta_2} \in F_{N_L-1}$. We also note that $\forall m \in \mathbb{Z}_+$ we can not reduce the F -degree of $\beta^m(\alpha\gamma^{\eta_1}\delta^{\eta_2})$ with any relations in S , hence it follows that $\bar{\beta}^m(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2}) \neq 0$. We now define right ideals $I_m := \bar{\beta}^m(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2})\text{gr}_F S$ for each $m \geq 1$, and consider the right ideal $I := \sum_{m \geq 1} I_m$. We claim that I is not finitely generated, which by definition implies that $\text{gr}_F S$ is not right Noetherian.

By Lemma 76 we have a grading $\text{gr}_F S = \bigoplus_{i \in \mathbb{N}} \bar{R}_i$ in the degree of $\bar{\beta}$, and note that we have shown above that $(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2})\bar{\beta} = 0$. Both of these facts together with the observation that the generators of I_m are homogenous with respect to the grading in the degree of $\bar{\beta}$ let us deduce that $\forall m \in \mathbb{Z}_+$, $I_m \subseteq \bar{R}_m$. This allows us to upgrade the sum in the

definition of I into a direct sum, i.e. we may write $I = \bigoplus_{m=1}^{\infty} I_m$.

Now, let $f_1, \dots, f_u \in I$ for some $u \in \mathbb{N}$, and define the right ideal $J := \sum_{j=1}^u f_j \text{gr}_F S$. We will show that $J \neq I$, by showing that $\exists M \in \mathbb{N}$ such that $J \subseteq \bigoplus_{m=1}^M I_m$. For each $j = 1, \dots, u$ we write $f_j = \sum_{l=1}^{M_j} h_{j_l}$ where $h_{j_l} \in I_l$ and $M_j \in \mathbb{N}$. Since there is a finite number of f_j we may define $M = \max\{M_j : 1 \leq j \leq u\}$. By construction of M and the grading of $\text{gr}_F S$ in the degree of $\bar{\beta}$, it follows that $\forall s_1, \dots, s_u \in \text{gr}_F S$, $\sum_{j=1}^u f_j s_j = \sum_{j=1}^u \sum_{l=1}^{M_j} h_{j_l} s_j \in \bigoplus_{m=1}^M I_m$, since $h_{j_l} s_j \in I_l$ for all j, l as $h_{j_l} \in I_l$ and I_l is a right ideal. This implies that $J \subseteq \bigoplus_{m=1}^M I_m \subset I$, where the strict inclusion follows since $\bar{\beta}^{M+1}(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2}) \in I$ as by definition of I we have $I_{M+1} \subseteq I$, but $\bar{\beta}^{M+1}(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2}) \notin J$ as $\bar{\beta}^{M+1}(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2}) \notin \bigoplus_{m=1}^M I_m$. Therefore, if I were to be finitely generated then we could show that the finite collection of generators does not generate I , a clear contradiction, hence it follows that I is not finitely generated.

The case $N_L < N_R$ follows mutatis mutandis from the argument above where we switch the roles of α and β . That is one can show that $(\bar{\beta}\bar{\gamma}^{\zeta_1}\bar{\delta}^{\zeta_2})\bar{\alpha} = 0$, define right ideals $I_m := \bar{\alpha}^m(\bar{\beta}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2})\text{gr}_F S$ for $m \geq 1$, and show that $I = \sum_m I_m$ is not finitely generated by considering the grading on $\text{gr}_F S$ in degree of $\bar{\alpha}$ and appropriately amending the above argument. The result now follows. $\square$

5.3 Future Research Directions

In this paper we have proved that for any monomial MS filtrations $F_{\bullet}$ on S the associated graded algebra $\text{gr}_F S$ is not Noetherian. The proof of Theorem 66 required the bigrading on $\text{gr}_F S$ as well as the fact that $(\bar{\alpha}\bar{\gamma}^{\eta_1}\bar{\delta}^{\eta_2})\bar{\beta} = 0$, however it is not clear why any of these two conditions should hold for more general MS filtrations, let alone more general filtrations. It is clear that another approach is needed. Perhaps a natural next step is to investigate non-monomial MS filtrations, and in order to do that one would need to come up with a classification of minimal generating sets of S . Another worthy endeavour would be to prove why S is minimally generated in four elements, or alternatively find a generating set with fewer generators. Both of these seem to be **hard** problems. Lastly, one could try to find ideals I, J_1, J_2 of S which would satisfy the hypothesis of Theorems 1.1 and 3.2 in [SZ03] to settle the MR question for S , or alternatively one could show that no such ideals exist. The work of this paper takes us a step closer to answering the MR question for the MS example S , and we are eager to see what new mathematics is developed in order to settle the MR question for S as well as the MR question in full generality.

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